

**ZEROTRUST AI AI-DRIVEN ACCESS MONITORING
AND SECURITY ANALYTICS PLATFORM**

PHASE I REPORT

Submitted by

MONISH D.Y.	230701195
MANICKA MEENAKSHI S.	230701173
MOHANAPRIYA R.	230701191
LAVANYA R.	230701162

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
ANNA UNIVERSITY: CHENNAI 600 025

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ANNA UNIVERSITY, CHENNAI

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“This project addresses the following Sustainable Development Goal(s):

SDG 8 – Decent Work and Economic Growth

SDG 9 – Industry, Innovation and Infrastructure

SDG 16 – Peace, Justice and Strong Institutions.”

SIGNATURE

Dr. J. Manoranjini,
Professor and Head,
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

SIGNATURE

Ms. K. Mahesmeena,
Assistant Professor
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

Submitted to Project Viva-Voce Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The rapid adoption of cloud computing, distributed applications, remote access, and heterogeneous devices has increased the complexity of modern cybersecurity environments and exposed the limitations of traditional perimeter-based security models. Conventional access-control mechanisms often rely on initial authentication and predefined trust, which may not adequately respond to changing user behaviour, anomalous activities, or policy violations. To address these challenges, this paper proposes an ML-driven Zero Trust security framework that integrates machine learning, Retrieval-Augmented Generation (RAG), Explainable Artificial Intelligence (XAI), and continuous security monitoring. The proposed framework follows the Zero Trust principle of continuous verification by evaluating each access request based on contextual and behavioural security information rather than granting implicit trust after authentication. Random Forest is employed for network threat classification, Isolation Forest is used to identify anomalous cloud-related activities, and Gradient Boosting is used for risk assessment. In addition, a RAG-based policy analysis mechanism retrieves relevant security policies from an external knowledge base and evaluates user activities against the retrieved policy context. XAI techniques are incorporated to improve the interpretability of machine learning-based security decisions by identifying the factors contributing to detected threats or elevated risk. The outputs from these components are integrated by a Zero Trust decision engine to dynamically classify requests as ALLOW, VERIFY, RESTRICT, or DENY. A controlled VirtualBox environment is used as the experimental platform for generating and analysing different security scenarios. The proposed framework aims to provide adaptive, policy-aware, explainable, and context-sensitive access control for modern distributed environments.

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MONISH D.Y. - 230701195

MANICKA MEENAKSHI S. - 230701173

MOHANAPRIYA R. – 230701191

LAVANYA R. - 230701162

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LIST OF ABBREVIATIONS

ACRONYM	EXPANSION
ZTA	Zero Trust Architecture
ML	Machine Learning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
RAG	Retrieval-Augmented Generation
MFA	Multi-Factor Authentication
UBA	User Behavior Analytics
UEBA	User and Entity Behavior Analytics
IDS	Intrusion Detection System
API	Application Programming Interface
RBAC	Role-Based Access Control
DDoS	Distributed Denial-of-Service
IP	Internet Protocol
VPN	Virtual Private Network
IAM	Identity and Access Management
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-Agnostic Explanations
MITRE	MITRE ATT&CK Knowledge Base
TTP	Tactics, Techniques, and Procedures
MFA	Multi-Factor Authentication
NLP	Natural Language Processing

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The rapid growth of cloud computing, web applications, remote access, and digital services has significantly increased the complexity of modern cybersecurity environments. Traditional security systems generally operate on the assumption that users or devices inside a trusted network can be granted access to resources with relatively limited verification. However, this assumption is increasingly unsuitable because modern applications are accessed from different devices, locations, networks, and cloud environments. Compromised credentials, malicious devices, insider threats, unauthorized access, and policy violations can therefore lead to serious security incidents.

Zero Trust Architecture (ZTA) addresses this limitation by following the principle of “never trust, always verify.” Instead of automatically trusting a user based on network location or previous authentication, Zero Trust continuously evaluates the identity, device, request, resource, context, and security posture before granting access. Access decisions are therefore made dynamically according to the current level of risk.

Machine Learning (ML) can strengthen Zero Trust security by identifying patterns associated with malicious or abnormal activities. Different machine learning models can be used for different security scenarios. In the proposed system, machine learning is applied to network activity, cloud-related behaviour, and Zero Trust risk assessment. Network activity can be analysed to identify potentially malicious traffic, while cloud activity can be examined for anomalous behaviour. A risk assessment model can further support dynamic access decisions.

Another important challenge is policy compliance. Security policies are often stored as documents containing rules regarding authentication, authorization, data access, user behaviour, and resource usage. Manually examining every user request against these policies is difficult and time-consuming. To address this problem, the proposed system incorporates a Retrieval-Augmented Generation (RAG) based policy-violation detection mechanism. The RAG component retrieves relevant policies from a knowledge base and uses the retrieved information to determine whether a particular activity or request complies with the defined security policies.

The system is also designed with a placement cell assistance module as a practical application scenario. Placement-related systems handle sensitive student and recruitment information, including eligibility details, applications, company requirements, and access to placement resources. The proposed system can provide controlled assistance while applying Zero Trust principles to protect sensitive information and ensure that access is consistent with organizational policies.

Therefore, the proposed work combines Zero Trust security, machine learning, anomaly/threat detection, RAG-based policy analysis, and placement-cell assistance into an integrated framework. The objective is not only to detect suspicious activities but also to use the detected risk and policy context to support appropriate access decisions.

1.2 OBJECTIVE

The primary objective of this project is to develop an ML-driven Zero Trust security framework capable of continuously evaluating users, devices, activities, and requests before allowing access to protected resources.

The major objectives of the proposed system are

1. To implement a Zero Trust security framework in which access is granted only after appropriate verification and risk evaluation.
2. To develop machine learning-based threat detection mechanisms for identifying suspicious or abnormal network and cloud activities.
3. To perform risk-based access assessment using machine learning so that security decisions can adapt to the current behaviour and context of a request.
4. To handle heterogeneous security datasets having different schemas and feature sets through appropriate preprocessing and feature-management techniques.
5. To develop a RAG-based policy-violation detection mechanism that retrieves relevant security policies and evaluates user activities or requests against them.
6. To provide explainable policy-related decisions by associating detected violations with the relevant policy information retrieved from the knowledge base.

7. To develop a placement cell assistance module that demonstrates how Zero Trust principles can be applied to a real-world student and recruitment environment.
8. To integrate security detection and access control into a unified decision mechanism that can classify requests into appropriate actions such as allowing, verifying, restricting, or denying access.
9. To evaluate the proposed framework through different security scenarios rather than relying only on conventional machine learning accuracy.
10. To demonstrate continuous security assessment, where a user's trust level is not considered permanent and can change according to new activities or detected risks.

1.3 EXISTING SYSTEM

Traditional cybersecurity systems commonly depend on perimeter-based security mechanisms. In such systems, users who successfully authenticate or users operating within a trusted network may receive access to multiple resources. This approach was more suitable when organizational resources were primarily located within a controlled internal network.

However, modern computing environments have changed considerably. Applications and data are distributed across cloud platforms, web applications, personal devices, remote networks, and third-party services. As a result, simply verifying a user's identity at the initial login stage is insufficient to guarantee that subsequent activities are legitimate.

Several limitations exist in conventional systems.

1. Static Trust

Traditional systems may assign trust based primarily on authentication credentials, network location, or predefined access roles. Once authenticated, a user may retain access even when their behaviour becomes suspicious.

2. Limited Continuous Verification

Many conventional systems perform authentication at login but do not continuously reassess the security condition of the user, device, or request.

3. Rule-Based Threat Detection

Traditional security mechanisms frequently depend on predefined signatures or rules. They may therefore have difficulty identifying previously unseen or behaviourally unusual attacks.

4. Separate Security Mechanisms

Network intrusion detection, cloud monitoring, access control, and policy enforcement are often implemented as separate components. The lack of integration can make it difficult to obtain a unified security decision.

5. Difficulty Handling Anomalies

Large-scale network and cloud environments generate substantial amounts of activity data. Manually identifying unusual behaviour from such data is inefficient.

6. Manual Policy Verification

Security and organizational policies may be available in document form. Checking every user activity manually against these policies is time-consuming and may lead to inconsistent decisions.

7. Lack of Context-Aware Access Decisions

Traditional access control mechanisms may not adequately consider factors such as current behaviour, device status, request context, detected threats, and policy violations simultaneously.

8. Limited Support for Dynamic Risk

A user's risk level can change over time. A user who was considered safe during one request may become suspicious after generating abnormal activity. Conventional static access mechanisms may not respond sufficiently quickly to such changes.

These limitations demonstrate the need for an integrated security architecture in which authentication, continuous monitoring, machine learning, policy compliance, and dynamic access control work together.

1.4 PROPOSED SYSTEM

The proposed system introduces an ML-driven Zero Trust framework integrated with RAG-based policy-violation detection and a placement cell assistance module.

The central principle of the proposed system is that no user or device is automatically trusted. Each access request is evaluated based on available identity, behavioural, network, cloud, and policy-related information.

The overall workflow is designed as follows:

User/Student → Application → Data Collection → Preprocessing → ML Analysis → Risk Assessment → RAG Policy Verification → Zero Trust Decision → Secure Resource Access

A. APPLICATION AND DATA COLLECTION

A controlled application environment is used to generate or process user and security-related activities. Relevant information can include network behaviour, cloud activity, user actions, device-related information, and access requests.

This provides a practical environment for demonstrating how security information moves through the proposed Zero Trust framework.

B. DATA PREPROCESSING AND SCHEMA HANDLING

The collected datasets may contain different feature structures, missing values, and inconsistent attributes. Therefore, preprocessing is performed before machine learning.

The preprocessing stage includes

- Missing-value handling
- Data cleaning
- Feature selection
- Data transformation
- Schema alignment
- Numerical feature processing
- Feature normalization where required

This stage enables heterogeneous security information to be used by the corresponding models.

C. MACHINE LEARNING-BASED SECURITY ANALYSIS

Different machine learning techniques are applied according to the characteristics of the security problem.

The proposed implementation includes:

- Random Forest for network-related threat/intrusion classification.
- Isolation Forest for identifying anomalous cloud activity.
- Gradient Boosting for Zero Trust risk assessment and classification.
- Rather than depending on a single model, the framework uses multiple security perspectives to obtain a more comprehensive understanding of a request.

D. RISK ASSESSMENT

The outputs of the security models are used as inputs to the risk evaluation layer.

For example, a request may have:

- Normal network behaviour
- Normal cloud activity
- Low behavioural risk and can therefore receive a lower risk assessment.

On the other hand, abnormal network activity, suspicious cloud behaviour, and an elevated risk score can cause the request to be subjected to additional verification or restriction.

E. RAG-BASED POLICY VIOLATION DETECTION

The proposed system incorporates a Retrieval-Augmented Generation architecture to handle policy-related decisions.

Security and organizational policies are maintained in a knowledge base. When a request or activity is received, the RAG system retrieves the most relevant policy information.

The retrieved policy context is then used to evaluate the activity.

For example:

User Request → Policy Retrieval → Relevant Policy → Policy Evaluation → Violation/Compliance Result

This approach is useful because the system does not have to rely solely on information embedded within the language model. Instead, it retrieves the relevant policy information from the organization's defined knowledge base.

The RAG layer can therefore assist in identifying:

- Unauthorized access attempts
- Access to restricted resources
- Policy-inconsistent activities
- Improper use of protected information
- Violations of predefined organizational rules

The retrieved policy information can also provide supporting context for the resulting decision.

F. ZERO TRUST DECISION ENGINE

The final decision is based on the security context associated with the request.

The decision engine can categorize a request into actions such as:

Risk / Policy Condition	Decision
Low risk + policy compliant	ALLOW
Moderate risk	VERIFY
Suspicious activity	RESTRICT
High risk / confirmed violation	DENY

Table 1.1. Decision Engine

This allows access decisions to be dynamic instead of permanently trusting a previously authenticated user.

G. PLACEMENT CELL ASSISTANCE MODULE

A practical application of the proposed framework is demonstrated through a placement cell assistance system. The placement environment can involve different users such as:

- Students
- Placement coordinators
- Recruiters
- Administrators

Each category requires different resources and permissions.

For example, a student may be permitted to access their own placement profile and eligible job opportunities, whereas an administrator may require access to broader placement information.

The Zero Trust layer ensures that access is evaluated according to the user's role, request context, security risk, and applicable policies.

The system can also assist with placement-related activities such as:

- Identifying eligible opportunities
- Matching student information with recruitment requirements
- Providing placement-related information
- Supporting administrative workflows
- Controlling access to sensitive placement information

Thus, the placement module provides a practical demonstration of how the proposed security framework can be applied beyond a purely theoretical cybersecurity environment.

H. CONTINUOUS MONITORING

Zero Trust does not consider trust to be permanent. After access is granted, subsequent activity can continue to be monitored. If the user's behaviour changes or a new threat is detected, the risk assessment can be updated and the access decision can be changed. For example:

ALLOW → suspicious behaviour detected → REASSESS → VERIFY / RESTRICT / DENY

This continuous evaluation is one of the major characteristics of the proposed framework.

The proposed system therefore combines:

Zero Trust Architecture + Machine Learning + RAG + Policy Compliance + Placement Assistance

to create a security framework capable of continuous verification, risk-based access control, threat detection, policy analysis, and secure application assistance. The proposed approach addresses several limitations of traditional perimeter-based and static access-control systems by making security decisions dynamic, context-aware, and supported by multiple sources of security information.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The increasing complexity of cloud-based applications, remote access environments, and distributed digital resources has resulted in the need for security mechanisms that continuously evaluate users, devices, and access requests. Existing research has explored several individual technologies including Zero Trust Architecture, machine learning-based intrusion detection, anomaly detection, Retrieval-Augmented Generation (RAG), and Explainable Artificial Intelligence (XAI). However, these technologies are often investigated independently. The proposed project combines these approaches into an integrated security framework.

2.1.1 ZERO TRUST ARCHITECTURE

[1] Rose et al. introduced the Zero Trust Architecture (ZTA) model through NIST Special Publication 800-207. The model shifts cybersecurity from traditional network-perimeter protection toward protection of individual users, devices, applications, and resources. It assumes that no implicit trust should be granted based solely on network location or ownership. Authentication and authorization are treated as separate security functions, and access decisions are made according to the current context of a request.

[2] NIST further emphasizes that Zero Trust environments should continuously inspect, monitor, and log interactions while applying risk-based access controls. This is particularly important in cloud and distributed environments where users may access resources from different devices and locations.

[3] NIST SP 800-207A further extends Zero Trust concepts to cloud-native and multi-cloud applications, emphasizing application and service identities and granular access policies rather than depending primarily on network parameters.

[4] Although Zero Trust provides a strong security architecture, the framework itself does not specify a single machine learning mechanism for determining behavioural risk. This creates an opportunity to integrate ML-based threat and anomaly detection into the Zero Trust decision process.

2.1.2 MACHINE LEARNING FOR INTRUSION DETECTION

Machine Learning has been extensively investigated for identifying malicious and anomalous network behaviour. Traditional intrusion detection systems commonly rely on predefined signatures and rules. Such approaches can be effective against known attacks but can have limitations when previously unseen or behaviourally different attacks occur. Research on ML-based intrusion detection has therefore explored classification and anomaly-detection approaches that learn patterns from network traffic. Recent research also emphasizes the importance of adapting intrusion detection systems to evolving threats, including changes in data distributions and feature behaviour. This indicates that security systems should not rely exclusively on static patterns. In the proposed system, Random Forest is used for network threat classification, while Isolation Forest is used for cloud-related anomaly detection. These models provide complementary views of security behaviour rather than depending on a single detection model.

2.1.3 MACHINE LEARNING-BASED RISK ASSESSMENT

Machine learning can also be used beyond simple attack classification. Security-related attributes such as user activity, network behaviour, device characteristics, and access context can be incorporated into a risk assessment mechanism. A risk-based approach is particularly suitable for Zero Trust because a security decision does not necessarily have to be binary. A request with low risk may be permitted, whereas a moderately suspicious request can be subjected to additional verification. A high-risk request can be restricted or denied. The proposed framework therefore incorporates Gradient Boosting for Zero Trust risk assessment. The resulting risk information can be combined with other security signals to support dynamic access decisions.

2.1.4 RETRIEVAL-AUGMENTED GENERATION FOR POLICY ANALYSIS

Large Language Models have demonstrated strong capabilities in understanding and generating natural language. However, directly asking a language model to make policy decisions without access to the latest organizational policies can result in incomplete or unreliable responses.

Retrieval-Augmented Generation (RAG) addresses this limitation by retrieving relevant information from an external knowledge source before generating an answer or decision. In a security-policy environment, organizational policies can be stored in a searchable knowledge base. When an activity or access request is received, the system can retrieve the policies relevant to that request.

The retrieved policy information can then be provided as context to the language model for policy analysis. In the proposed framework, the RAG component is used as a policy compliance layer:

User Activity → Policy Query → Relevant Policy Retrieval → Policy Analysis → Compliance/Violation.

This approach allows the system to evaluate activities against defined policies rather than relying exclusively on the language model's general knowledge.

2.1.5 EXPLAINABLE ARTIFICIAL INTELLIGENCE IN CYBERSECURITY

Machine learning models can produce highly useful predictions, but some models may be difficult for humans to interpret. This creates a significant challenge in cybersecurity because a security administrator may need to understand why an activity was classified as malicious or why an access request received a high-risk score. Research has highlighted the importance of Explainable AI for cybersecurity because security analysts often need understandable explanations for automated alerts and predictions. XAI can help reduce the difficulty of interpreting complex model decisions and can support investigation and trust in automated security systems. Recent research specifically examining XAI for intrusion detection identifies techniques such as SHAP and LIME as important approaches for explaining machine-learning-based security predictions. SHAP can identify the contribution of individual features toward a prediction, while LIME can construct a local interpretable approximation around an individual prediction. These explanations can help a security analyst understand which factors contributed to a threat or risk classification.

Therefore, the proposed project incorporates an Explainable AI layer alongside the ML models.

For example:

- Prediction: High Risk
- Explanation: Unusual network behaviour, elevated request frequency, and abnormal cloud activity contributed significantly to the risk score.

This transforms the system from simply providing a prediction into a system capable of providing an understandable justification for that prediction.

2.1.6 EXPLAINABLE AI AND INTRUSION DETECTION

Recent studies specifically investigate the combination of ML and XAI for network intrusion detection. Research has shown that combining detection models with explanation techniques can improve the transparency of security predictions and help security analysts understand the factors influencing an alert. Other recent work has also emphasized that explainability is important because black-box security decisions can reduce user confidence and make investigation more difficult. However, explainability itself introduces challenges, including computational cost, explanation stability, and potential security risks. Therefore, XAI should be treated as a supporting component of the security framework rather than as a replacement for the underlying detection models.

2.1.7 INTEGRATION OF ZERO TRUST, ML, RAG AND XAI

The reviewed literature demonstrates that each technology addresses a different aspect of cybersecurity:

- Zero Trust addresses dynamic access control and removal of implicit trust.
- Machine Learning identifies malicious or anomalous behaviour.
- RAG provides policy-aware analysis using an external knowledge base.
- Explainable AI provides human-understandable reasoning for ML predictions.

However, these capabilities are commonly addressed as separate research problems. The proposed project attempts to integrate them into a unified framework.

The proposed architecture therefore follows:

Zero Trust → ML Detection → Risk Assessment → XAI Explanation → RAG Policy Verification → Final Access Decision

This integration enables the system to answer not only:

- “Should access be granted?”
- but also:
- “What is the risk?”
- “Why was this risk assigned?”
- “Does the request violate any defined policy?”
- “What action should be taken?”

This provides a more comprehensive approach to security decision-making.

2.2 COMPARISON AND DISCUSSION

Traditional security mainly relies on predefined rules, while ML improves abnormal-behaviour detection. Zero Trust enables risk-based access control, RAG supports policy-aware analysis, and XAI improves transparency. The proposed framework integrates these capabilities to make **behaviour-, risk-, policy-, and evidence-based security decisions**.

It also supports adaptive responses: **low-risk requests can be allowed, moderate-risk requests can require verification, and high-risk or policy-violating requests can be restricted or denied**. In the placement cell environment, this enables secure and role-specific access to student, recruiter, and placement information.

Additionally, the framework enhances **security consistency and decision accuracy** by combining multiple security mechanisms instead of relying on a single detection method. This integrated approach helps reduce unauthorized access and provides **more reliable and context-aware security decisions**.

The major characteristics of these approaches are summarized in below table:

Approach	Threat Detection	Zero Trust	Policy Analysis	Explainability	Dynamic Decision
Traditional Rule-Based Security	Limited to known rules	✗	Rule-based	Limited	Limited
Conventional ML-IDS	✓	✗	✗	Usually limited	Partially
Zero Trust Architecture	✗	✓	✓ through policies	Limited	✓
ML-Based Anomaly Detection	✓	✗	✗	Depends on model	✓
RAG-Based Policy Analysis	✗	✗	✓	Context-based	Partially
XAI-Based Security	✓	✗	✗	✓	Depends on model
Proposed Framework	✓	✓	✓ RAG	✓ XAI	✓

Table 2.1. Comparison Table

2.3 CONCLUSION

The literature survey highlights **Zero Trust Architecture** for continuous verification and secure access control. **Machine learning** detects abnormal behaviour, while **Explainable AI** improves decision transparency. **RAG** enables policy-based security decisions using an updateable knowledge base. The proposed project integrates **Zero Trust, Machine Learning, RAG, and Explainable AI** into a unified framework for **threat detection, risk assessment, policy-violation detection, and secure placement assistance**. This combination forms the foundation for the system design and implementation in the following chapters.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed system is an integrated Zero Trust security framework that combines Machine Learning, Retrieval-Augmented Generation (RAG), Explainable Artificial Intelligence (XAI), and a placement cell assistance module. The system follows the principle of continuous verification, where every user request is evaluated based on identity, behaviour, context, security risk, and policy compliance before access is granted.

The system starts when a user interacts with the application. Users may include students, placement coordinators, recruiters, and administrators. Authentication establishes the user's identity, but access is not automatically granted. The request is further analysed using network, cloud, behavioural, and contextual information.

The collected data is passed through a preprocessing and schema-management layer. This layer handles missing values, feature transformation, data types, and differences between the feature structures required by the various ML models. It prepares the appropriate input for network threat detection, cloud anomaly detection, and Zero Trust risk assessment.

The ML layer consists of multiple models. Random Forest is used for network threat classification, Isolation Forest is used for detecting abnormal cloud behaviour, and Gradient Boosting is used for Zero Trust risk assessment. The outputs of these models provide security signals that contribute to determining the overall risk associated with a request.

The RAG component provides policy-aware analysis. Security and placement policies are stored in a knowledge base. When a request requires policy verification, the system retrieves the relevant policy information and provides it as context to the language model. The system can therefore determine whether the activity is compliant with the organization's defined policies or represents a potential violation.

The Explainable AI layer improves the transparency of ML predictions. Techniques such as SHAP or LIME can be used to identify the important features influencing a prediction. Instead of displaying only a high-risk result, the system can provide an understandable explanation of the factors that contributed to the decision.

The Zero Trust decision engine combines the risk assessment, policy result, user context, and other security information. Based on the overall condition, the system can allow the request, require additional verification, restrict the activity, or deny access. After access is granted, the user's subsequent activity continues to be monitored so that the security decision can be reassessed when required.

The placement cell module provides the practical application environment. Students can access authorized placement information and opportunities, while coordinators and recruiters receive access according to their respective permissions. The Zero Trust framework ensures that sensitive placement information is not automatically accessible merely because a user has successfully authenticated.

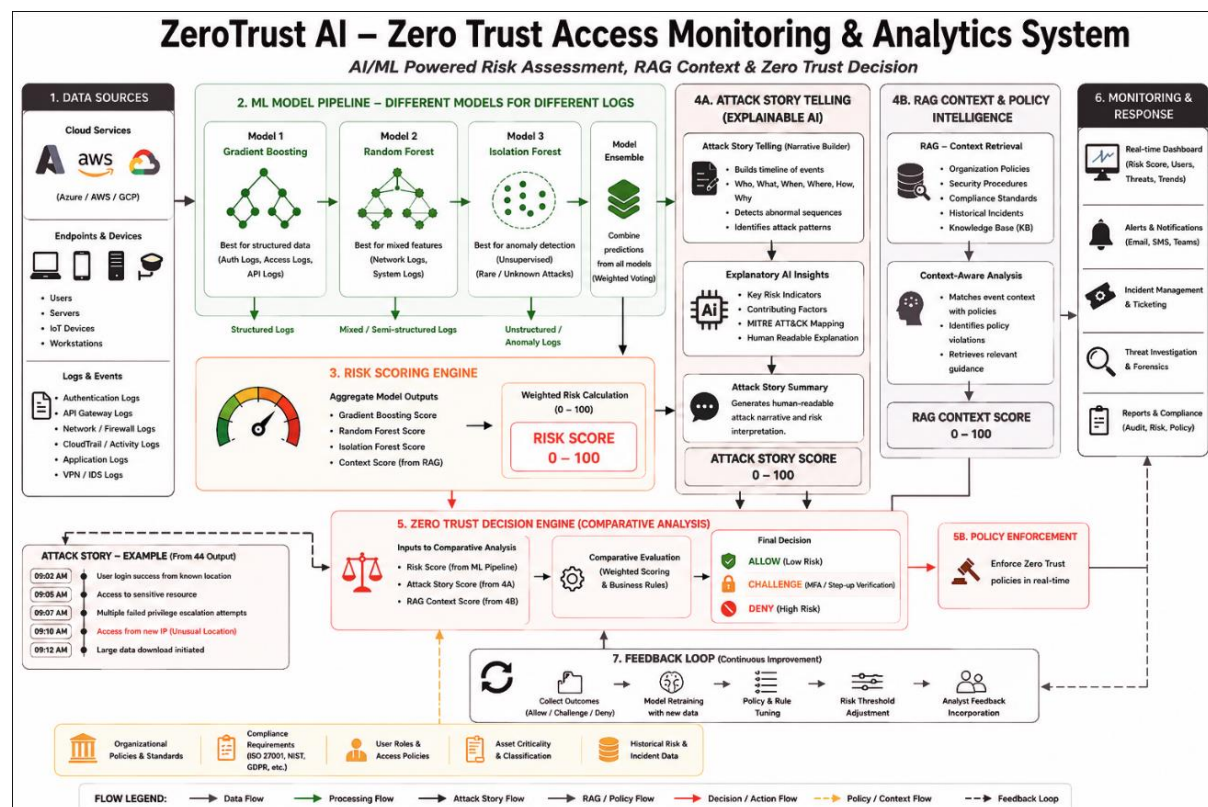


Figure 3.1. Architecture Diagram of the system

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

The proposed system can be implemented using Python as the primary programming language. Pandas and NumPy are used for data processing, while Scikit-learn is used for implementing the Random Forest, Isolation Forest, and Gradient Boosting models. SHAP or LIME can be integrated for Explainable AI.

The RAG component requires a document-processing mechanism, embedding model, vector database or similarity-search system, and a suitable language model. Flask or FastAPI can be used for backend development, while HTML, CSS, and JavaScript can be used for the application interface. Jupyter Notebook or Visual Studio Code can be used for development and experimentation.

3.2.2 HARDWARE REQUIREMENTS

The system can be developed on a standard computer with an Intel Core i5 or equivalent processor, at least 8 GB RAM, and approximately 256 GB of storage. A GPU is not mandatory for the traditional machine learning models. Additional computational resources may be useful if a large language model is executed locally for the RAG component.

3.2.3 DATASET REQUIREMENTS

The proposed system uses different categories of data for its security components. Network security data is used for intrusion and threat classification, while cloud activity data is used for anomaly detection. Behavioural and contextual data is used by the Zero Trust risk-assessment model.

The RAG component requires a policy knowledge base containing security, privacy, access-control, and placement-related policies. Sample or synthetic placement data can be used to demonstrate student information, company requirements, eligibility, and placement activities without exposing real confidential information.

3.3 SYSTEM DESIGN

The system design defines how the application, security models, RAG component, Explainable AI, and Zero Trust decision engine interact with one another. Every protected request passes through the security framework before the requested resource is provided.

3.3.1 ACTIVITY DIAGRAM

The activity begins when a user submits a request through the application. The system authenticates the user and collects the relevant contextual information. The data is then preprocessed and supplied to the appropriate machine learning models for threat and risk analysis. The RAG component retrieves the relevant policy whenever policy validation is required. At the same time, the Explainable AI component provides an understandable interpretation of the ML prediction. The resulting risk, policy status, and contextual information are then evaluated by the Zero Trust decision engine. The request is finally classified as ALLOW, VERIFY, RESTRICT, or DENY. If access is granted, the user's subsequent activity is monitored and the security assessment can be repeated whenever suspicious behaviour is detected.

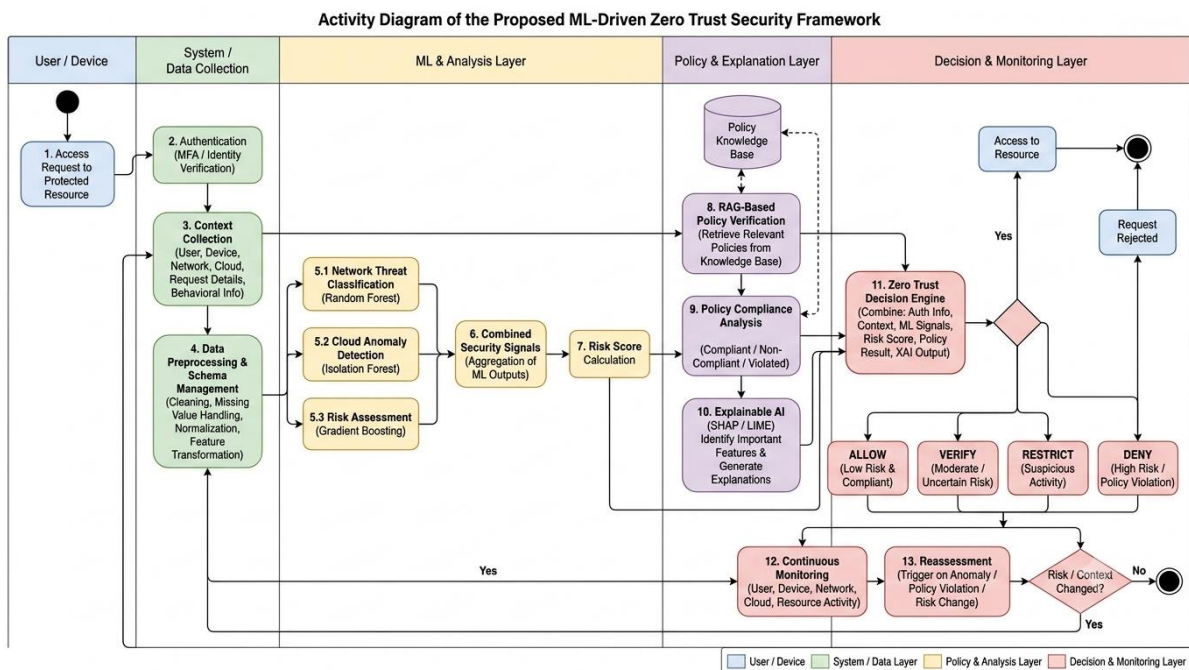


Figure 3.2. Activity Diagram

3.3.2 DATA FLOW DIAGRAM

The data flow begins with the user and placement application and proceeds to the data collection and preprocessing layers. The processed information is then distributed to the Random Forest, Isolation Forest, and Gradient Boosting models according to the type of security analysis required.

The model outputs are combined during risk assessment. The activity is also passed to the RAG component when policy verification is required, while the XAI layer generates an explanation for the relevant ML prediction. These outputs are provided to the Zero Trust decision engine, which determines the final access action.

If access is allowed, the placement application provides the requested functionality. Otherwise, the system performs additional verification or restricts the request. Continuous monitoring ensures that subsequent activities can trigger a new assessment when the user's security condition changes.

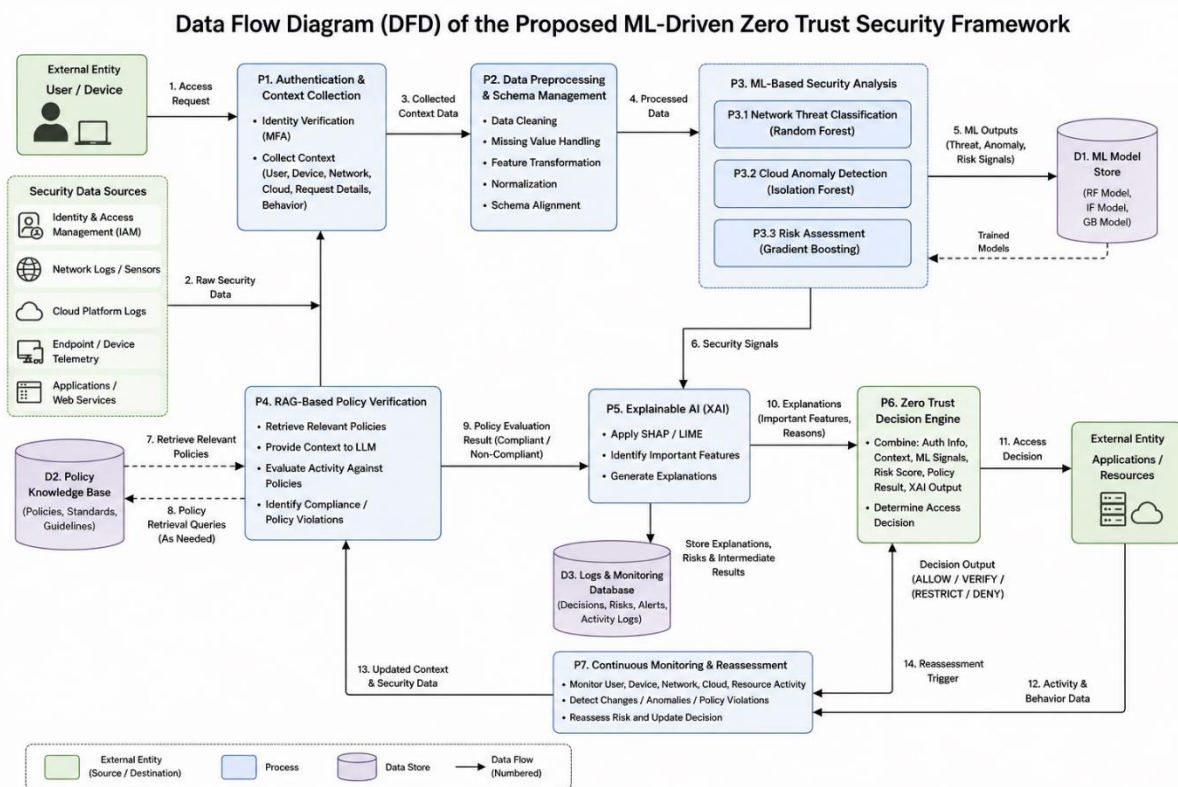


Figure 3.3. Data Flow Diagram

CHAPTER 4

MODULE DESCRIPTION

4.1 METHODOLOGIES

The proposed project follows an integrated methodology for developing a secure and intelligent application based on Zero Trust principles. The methodology combines data preprocessing, machine learning, Retrieval-Augmented Generation, Explainable Artificial Intelligence, policy compliance analysis, and continuous security monitoring. The main objective of this methodology is to ensure that access decisions are not based only on user authentication but are determined by the current security condition of the user, device, request, and requested resource.

The development process begins with the collection and preparation of security-related datasets. Since the proposed framework uses multiple machine learning models, the datasets may contain different schemas and feature sets. The data is therefore cleaned and transformed before model training. Missing attributes are handled appropriately, irrelevant features are removed, and the remaining features are transformed into the format expected by the corresponding models. This ensures that inconsistent input structures do not directly affect the security predictions.

Following preprocessing, separate machine learning models are trained for different security requirements. Random Forest is used for network threat classification, Isolation Forest is used for detecting anomalous cloud behaviour, and Gradient Boosting is used for Zero Trust risk assessment. The models are evaluated using appropriate classification and anomaly-detection metrics, while scenario-based testing is also performed to determine how the models behave under different security conditions.

The next stage incorporates the RAG mechanism for policy analysis. Security and organizational policies are converted into a searchable knowledge base. When a user performs an activity that requires policy validation, the system retrieves the relevant policy information and provides it as context to the language model. The generated response is then used to determine whether the activity is compliant with the applicable policy or represents a violation.

Explainable Artificial Intelligence is incorporated to make the machine learning predictions understandable. The explanation mechanism analyses the factors contributing to the model's prediction and presents the important features associated with the detected risk. This is particularly useful when a request is restricted or denied because it allows the administrator to understand the reason behind the decision.

Finally, the outputs from the machine learning models, policy analysis, contextual information, and explanations are provided to the Zero Trust decision mechanism. The system determines whether the request should be allowed, verified, restricted, or denied. The resulting activity is continuously monitored so that a change in user behaviour can trigger a new security assessment.

4.2 MODULES

The proposed system is divided into several interconnected modules so that data processing, security analysis, policy verification, decision-making, and application functionality can be developed and maintained independently. The modules communicate with one another through the defined data flow, while the Zero Trust decision mechanism acts as the central security-control component.

4.2.1 DATA MANAGER PACKAGE

The Data Manager Package is responsible for handling the security and application data required by the proposed framework. It acts as the initial processing layer between the application and the machine learning components. The package manages the loading, organization, validation, preprocessing, and transformation of data before it is supplied to the corresponding security models.

The package handles multiple datasets because the proposed system contains separate models for network threat detection, cloud anomaly detection, and Zero Trust risk assessment. Each dataset can contain a different number of attributes and a different feature structure. Therefore, the Data Manager Package identifies the schema associated with each model and prepares the incoming data accordingly.

Missing features are handled during this stage so that the model receives the required input structure. The package also performs data-type conversion, categorical encoding, numerical processing, feature selection, and other preprocessing operations required by the trained models. The same preprocessing logic used during model development is maintained during prediction to avoid inconsistencies between training and real-time inputs.

The package additionally manages the policy documents required by the RAG component. Policy documents are loaded, cleaned, divided into appropriate text segments, and converted into representations suitable for retrieval. This allows the policy knowledge base to be updated independently from the machine learning components.

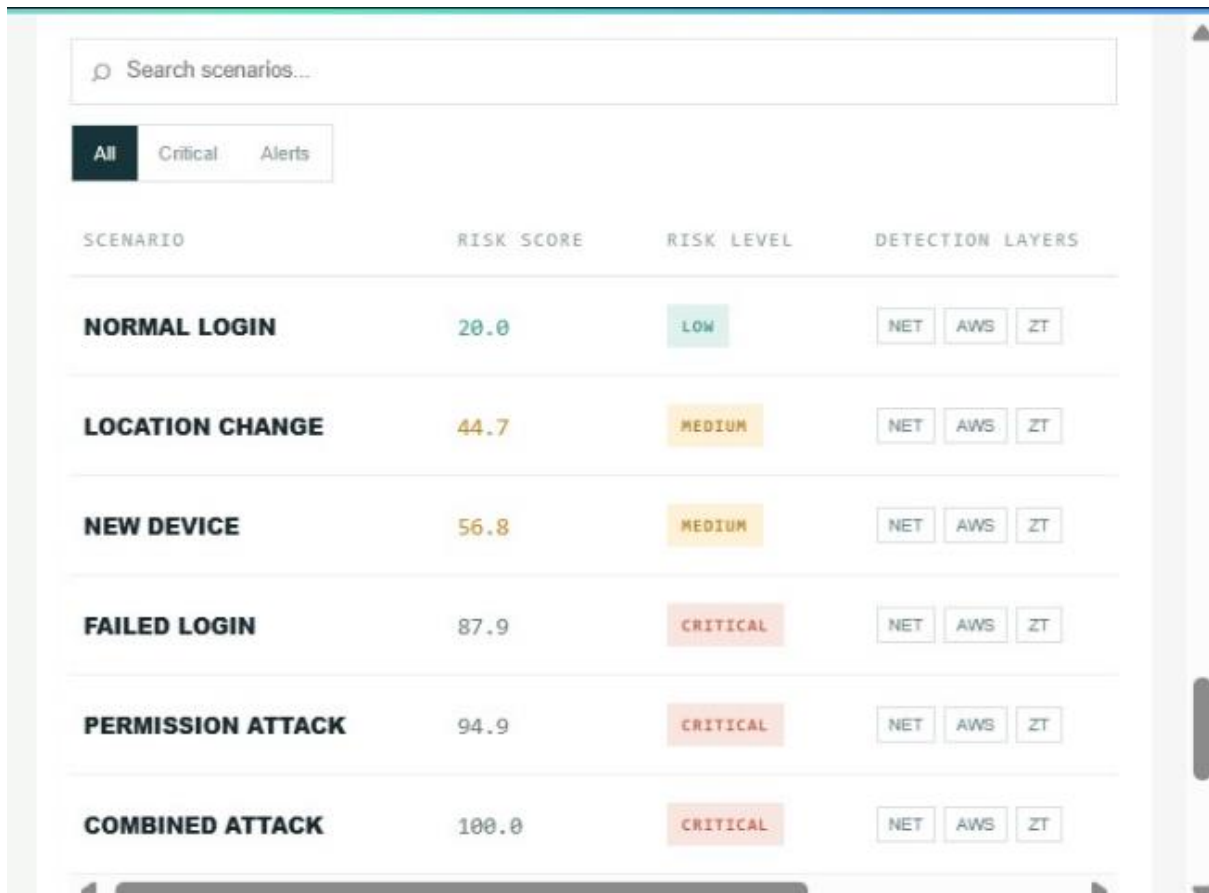
4.2.2 CORE PACKAGE

The Core Package represents the central security-processing component of the proposed system. It coordinates the interaction between the incoming request, machine learning models, risk assessment mechanism, policy analysis component, and Zero Trust decision engine.

When a request is received, the Core Package obtains the relevant contextual information and forwards it to the Data Manager Package for preprocessing. The processed information is then supplied to the appropriate machine learning models. The Random Forest model analyses network-related activity, while the Isolation Forest model examines cloud-related behaviour. The Gradient Boosting model evaluates the relevant Zero Trust attributes and contributes to the overall risk assessment.

The Core Package collects the outputs of these models and converts them into security signals. These signals are combined with information such as the user's role, requested resource, current activity, and policy status. The resulting information is passed to the Zero Trust decision engine, which determines the appropriate action for the request.

The Core Package also coordinates the Explainable AI mechanism. When a machine learning prediction is generated, the corresponding explanation is produced so that the security administrator can understand the factors that influenced the prediction. This ensures that the core security mechanism provides both a decision and supporting information.



SCENARIO	RISK SCORE	RISK LEVEL	DETECTION LAYERS
NORMAL LOGIN	20.0	LOW	NET AWS ZT
LOCATION CHANGE	44.7	MEDIUM	NET AWS ZT
NEW DEVICE	56.8	MEDIUM	NET AWS ZT
FAILED LOGIN	87.9	CRITICAL	NET AWS ZT
PERMISSION ATTACK	94.9	CRITICAL	NET AWS ZT
COMBINED ATTACK	100.0	CRITICAL	NET AWS ZT

Figure 4.1. Core Package

4.2.3 MODERATION PACKAGE

The Moderation Package is responsible for policy and activity compliance analysis. It integrates the RAG-based policy-violation detection mechanism into the security framework. The package is designed to determine whether a user's activity is consistent with the organization's defined policies.

When an activity requires policy verification, the Moderation Package converts the activity into a suitable query and searches the policy knowledge base. Relevant policy sections are retrieved using semantic similarity. The retrieved information is then provided to the language model as contextual information for policy evaluation.

The language model analyses the activity against the retrieved policy and generates a structured compliance result. The result can indicate whether the activity is compliant, potentially suspicious, or a policy violation. The relevant policy information is retained as supporting evidence for the decision.

For example, if a student attempts to access information belonging to another student, the Moderation Package can retrieve the relevant confidentiality and access-control policies. If the requested operation is inconsistent with those policies, the system identifies the activity as a potential policy violation and forwards the result to the Zero Trust decision engine.

This package therefore provides an additional security layer beyond machine learning. Even if an activity does not appear malicious from a network perspective, it can still be restricted when it violates an organizational policy.

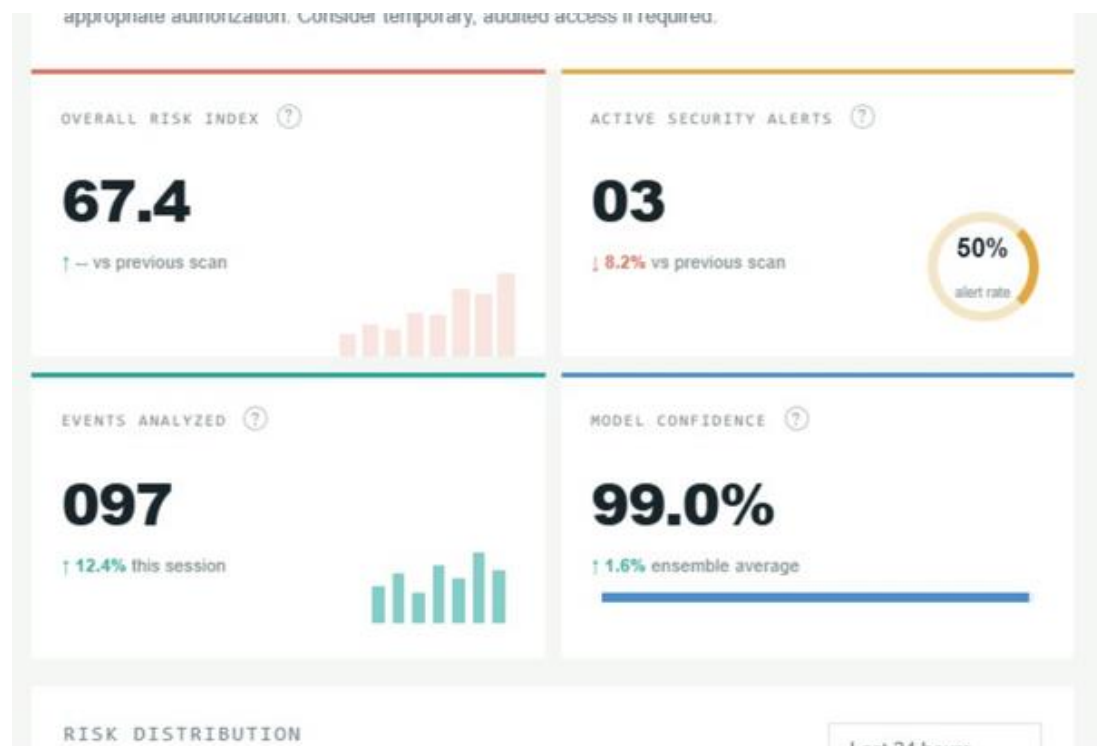


Figure 4.2. Risk Score

4.2.4 REINFORCEMENT PACKAGE

The Reinforcement Package represents the continuous decision and feedback component of the proposed framework. It is responsible for maintaining the relationship between security events, risk assessments, access decisions, and subsequent user behaviour.

The package does not treat an access decision as permanent. After a request is allowed, subsequent activities continue to be observed. If the system detects a significant change in behaviour, the new activity is again evaluated through the security pipeline. This allows the user's security status to change according to the latest available evidence.

The package can use the outcome of previous security decisions as feedback for subsequent assessments. For example, repeated suspicious requests, increasing anomaly scores, or repeated policy violations can contribute to a higher risk state. Conversely, consistently normal behaviour can provide evidence of a stable security condition.

This continuous feedback mechanism supports the fundamental Zero Trust principle that trust should be continuously evaluated. The system can consequently move a user from an ALLOW state to VERIFY, RESTRICT, or DENY when new evidence indicates an increased security risk.

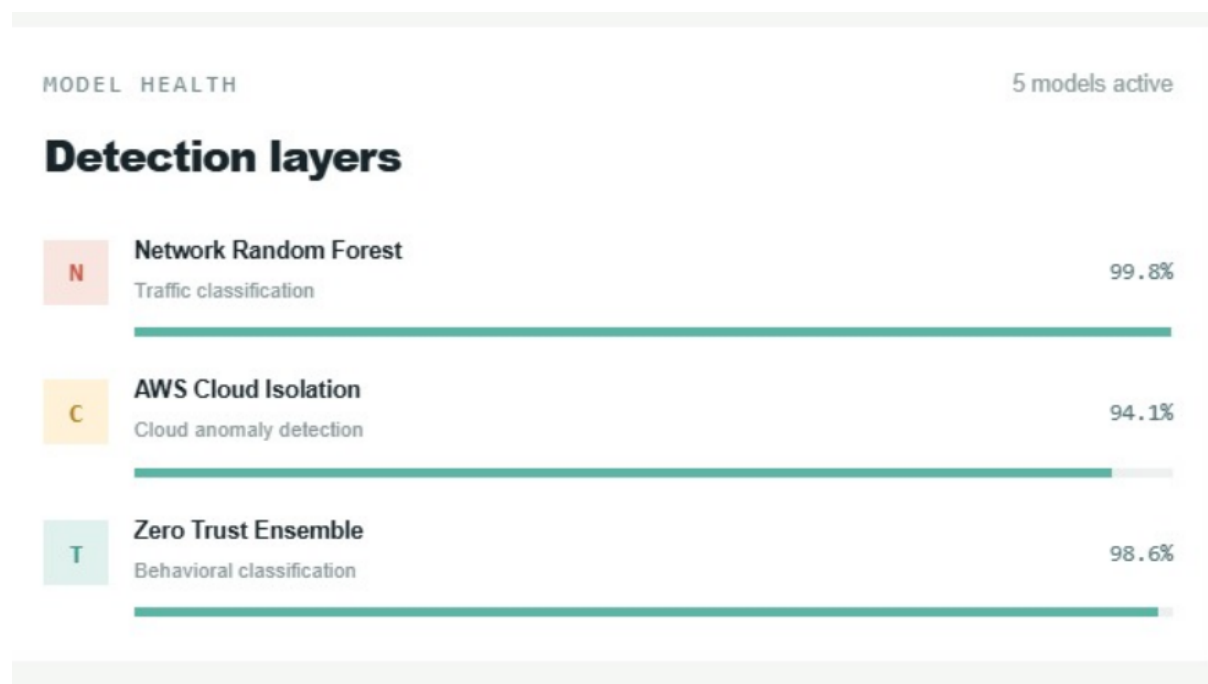


Figure 4.3. Detection Layers

4.2.5 UI PACKAGE

The UI Package provides the interface through which users and administrators interact with the proposed system. The interface is designed to provide access to the placement-cell functionality while integrating the security decisions generated by the Zero Trust framework.

Students can use the interface to access their authorized placement information, view relevant opportunities, and receive assistance based on their eligibility and available placement information. The system can compare relevant student attributes with recruitment requirements and provide appropriate placement-related assistance without exposing unauthorized information.

Placement coordinators and administrators can access functions according to their authorized permissions. The interface can display relevant security information when an access request is considered suspicious. Administrators can also view the risk status and Explainable AI output associated with a security decision.

The UI Package communicates with the backend security components whenever a protected operation is requested. The user therefore does not directly access sensitive resources without passing through the Zero Trust decision mechanism. If the request is permitted, the requested functionality is provided.

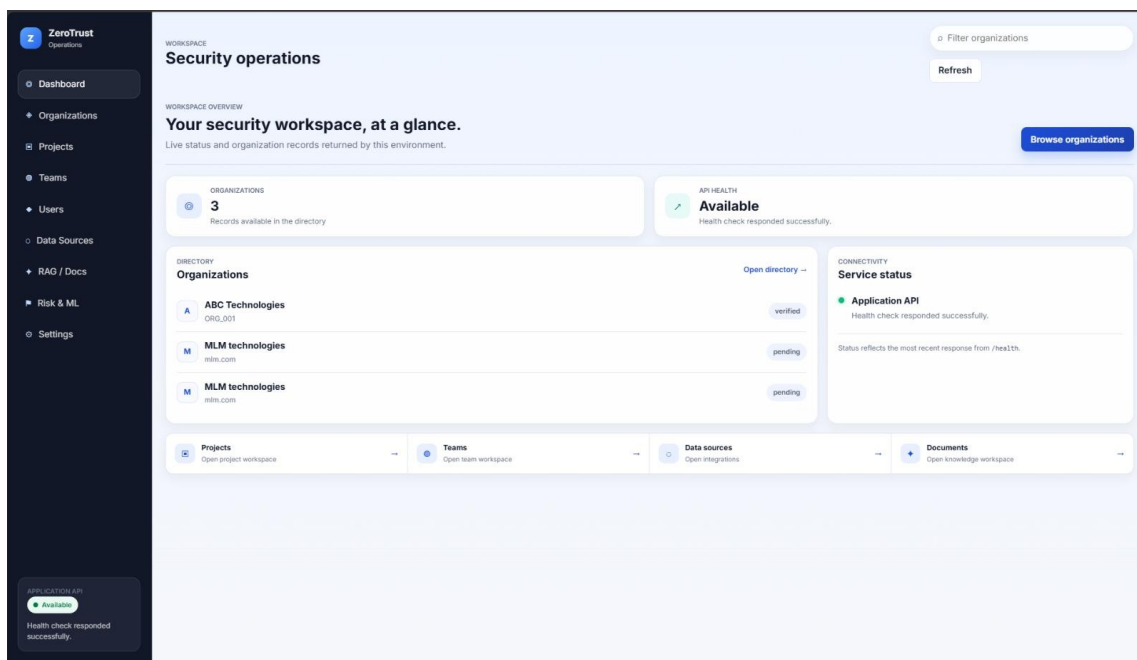
If additional verification is required, the interface can request the necessary verification. When the request is restricted or denied, the interface can display an appropriate security response.

The UI therefore acts as the interaction layer between the user and the integrated security framework while ensuring that application functionality remains subject to Zero Trust policies.

Overall Module Interaction:

- The modules work together as a continuous security pipeline. The UI Package receives a user request and forwards the required information to the Core Package. The Data Manager Package prepares the relevant data and supplies it to the machine learning models. The resulting predictions are processed by the Core Package to determine the current risk condition.

- Whenever policy validation is required, the Moderation Package retrieves the relevant policy through the RAG mechanism and determines whether the activity complies with the defined rules. The Explainable AI component provides an interpretable explanation of the machine learning result. The Core Package then combines the risk information, policy result, contextual information, and security explanation before forwarding them to the Zero Trust decision mechanism.
- The Reinforcement Package maintains continuous feedback from subsequent activities so that the security state can be reassessed over time. The final decision is communicated back to the UI Package, which provides or restricts the requested application functionality.
- This modular design allows the individual components to be developed, tested, modified, and scaled independently while maintaining a unified Zero Trust security framework.



ZeroTrust

Operations

- Dashboard
- Organizations
- Projects
- Teams
- Users
- Data Sources
- RAG / Docs
- Risk & ML
- Settings

Health check on

Available

Health check responded successfully.

WORKSPACE

Organizations

DISCOVERY

Organizations

View and manage organizations registered in this workspace.

New organization

Search by name, domain, or ID

Refresh

Organization directory

Organization details are loaded from the application API.

3 organizations

Directory records loaded from the API.

ORGANIZATION	DOMAIN	STATUS
ABC Technologies ORG_001	Not provided	verified
MLM technologies ORG_000001	mlm.com	pending
MLM technologies ORG_000002	mlm.com	pending

ZeroTrust

Operations

- Dashboard
- Organizations
- Projects
- Teams
- Users
- Data Sources
- RAG / Docs
- Risk & ML
- Settings

Health check on

Available

Health check responded successfully.

WORKSPACE

Projects

PORTFOLIO

Projects

Create project

Organization

ABC Technologies

Refresh

Sign out

3 projects loaded.

ZEROTRACE

vanakkam da mapla

0 members

active

ZEROTRACE

vanakkam da mapla

0 members

active

ZEROTRACE

vanakkam da mapla

0 members

active

ZeroTrust

Operations

- Dashboard
- Organizations
- Projects
- Teams
- Users
- Data Sources
- RAG / Docs
- Risk & ML
- Settings

Health check on

Available

Health check responded successfully.

WORKSPACE

Teams

GROUPS

Teams

Add team

Refresh

3 teams loaded.

Backend Team

Backend development team

1 members active

SOC Incident Response

Security Operations Center

0 members active

Unlabeled

ZeroTrace

2 members active

ZeroTrust

Operations

Dashboard

Organizations

Projects

Teams

Users

Data Sources

RAG / Docs

Risk & ML

Settings

Health check responded successfully

WORKSPACE

Users

PEOPLE

Users

Search users

Refresh

5 of 5 users.

Admin User

admin@abc.com · Organization Admin

active

Meenakshi

meenakshi@abc.com · Developer

active

string1

user@example1.com · No role assigned

invited

User UI Check 20261004

user.ui.check.20261004@example.com · No role assigned

invited

Lavanya

23070162@rajalakshmi.edu.in · No role assigned

invited

ZeroTrust

Operations

Dashboard

Organizations

Projects

Teams

Users

Data Sources

RAG / Docs

Risk & ML

Settings

Health check responded successfully

WORKSPACE

RAG / documents

PROVIDER

RAG / Documents

Refresh

3 documents loaded.

Upload PDF

PDF document

Choose File No file chosen

Upload document

Ask documents

Question

Why am I unauthorized

Results

5

Ask

Answer

For the question "Why am I unauthorized", the indexed compliance evidence says: 57cc5c86-1451-4126-90c1-258f0a228f4.pdf, page 12: which is reflected in the fact that the dynamic rules preserve the existing rules or generates new stricter rules for malicious users who are unable to access the system in Fig. 12a. In Fig. 13b, user 12 obtains the permission "compute : start" in the online launch phase, and the trust value continuously increases since user 12 has been behaving well until epochs 180. However, the trust assessment methodology tracks malicious behaviors during epochs 180-250, and the trust value drops significantly in the aforementioned epochs. In Fig. 13c, the static rules maintain authorization to user 12 after epochs 250 since the current trust value still satisfies the policy rule that is higher than the trust threshold (0.75), and the user may perform more malicious behaviors until the trust value 57cc5c86-1451-4126-90c1-258f0a228f4.pdf, page 11: ning the constant request action "compute : start" to evaluate the performance changes of dynamic authorization and permission transitions before and after the introduction of dynamic rules. (1) user 1 with historical trust 0.8 securely executes all requested permissions. (2) user 4 with historical trust 0.1 maliciously executes all requested permissions during training. (3) user 12 with historical trust 0.7 executes malicious operations that violate permissions for a period of time but safely executes all requests the rest of the time. (4) user 36 with historical trust 0.7 always performs malicious operations outside the requested permissions at a low frequency. The following experiments analyze the first 1000 epochs of Tonline . As we can observe from Fig. 11c, static rules Fig. 8 DQ

Keep access restricted unless the user has an approved administrative role. If access is required, use approved temporary privileged access with authorization and logging.

which is reflected in the fact that the dynamic rules preserve the existing rules or generates new stricter rules for malicious users who are unable to access the system in Fig. 12a. In Fig. 13b, user 12 obtains the permission "compute : start" in the online launch phase, and the trust value continuously increases since user 12 has been behaving well until epochs 180. However, the trust assessment methodology tracks malicious behaviors during epochs 180-250, and the trust value drops significantly in the aforementioned epochs. In Fig. 13c, the static rules maintain authorization to user 12 after epochs 250 since the current trust value still satisfies the policy rule that is higher than the trust threshold (0.75), and the user may perform more malicious behaviors until the trust value

ZeroTrust

Operations

Dashboard

Organizations

Projects

Teams

Users

Data Sources

RAG / Docs

Risk & ML

Settings

Health check responded successfully

WORKSPACE

Data sources

INTEGRATIONS

Data Sources

Refresh

43 data sources loaded.

Data Source UI Check 20261004

Api

Temporary frontend integration check

active

Enabled

Test connection

Security Event Stream

Stream

KAFKA stream

CONNECTED

Enabled

Test connection

Authentication Logs

File

Uploaded file authentication_logs.csv

READY

Enabled

Test connection

Internal SQLite

Database

SQLite database connection

CONNECTED

Enabled

Test connection

Customer PostgreSQL

Database

POSTGRESQL database connection

CONNECTED

Enabled

Test connection

Customer Webhook

Webhook

WEBHOOK source

READY

Enabled

Test connection

Customer Security API

Api

API source

READY

Enabled

Test connection

Security Event Stream

Stream

KAFKA stream

CONNECTED

Enabled

Test connection

Authentication Logs

File

Uploaded file authentication_logs.csv

READY

Enabled

Test connection

Internal SQLite

Database

SQLite database connection

CONNECTED

Enabled

Test connection

Customer PostgreSQL

Database

POSTGRESQL database connection

CONNECTED

Enabled

Test connection

Customer Webhook

Webhook

WEBHOOK source

READY

Enabled

Test connection

30

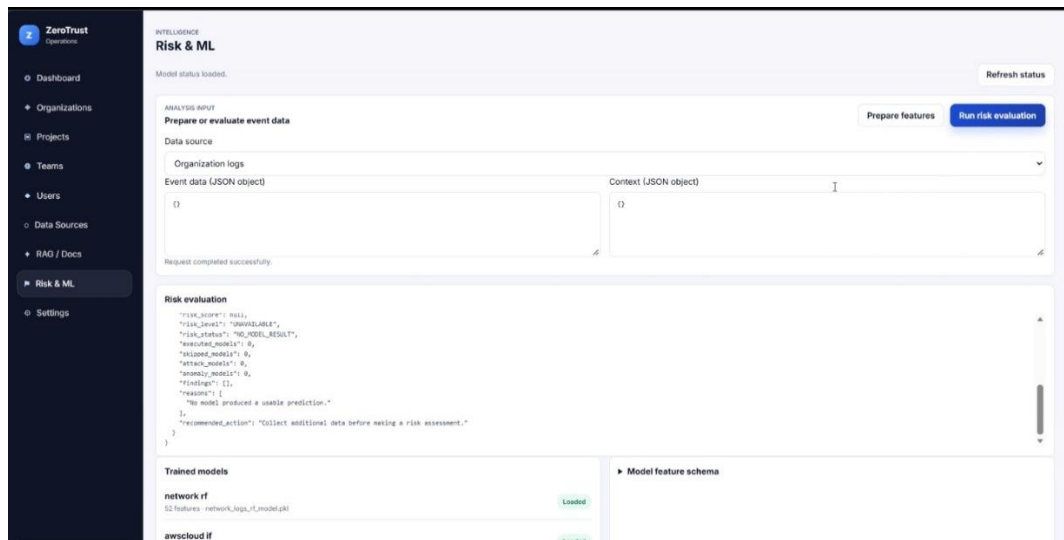


Figure 4.4. UI

CHAPTER 5

IMPLEMENTATION

5.1 IMPLEMENTATION

The proposed system is implemented as an integrated security framework combining Zero Trust Architecture, machine learning, Retrieval-Augmented Generation, Explainable Artificial Intelligence, and placement cell assistance. The implementation is divided into different modules so that each security function can be developed and evaluated independently while contributing to the final access-control decision.

The implementation begins with the preparation of security datasets. The network, cloud, and Zero Trust datasets are loaded and examined for missing values, inconsistent attributes, duplicate records, and different data types. Since each dataset contains a different number of features, schema handling is performed to ensure that the input provided to each trained model matches its expected structure.

The preprocessed data is then provided to the machine learning models. The Random Forest model is implemented for network threat detection, the Isolation Forest model is used for identifying anomalous cloud behaviour, and the Gradient Boosting model is used for Zero Trust risk assessment. The trained models are stored and loaded when a new security request needs to be evaluated.

During runtime, the application collects information associated with a user's request and passes the required attributes through the preprocessing pipeline. The corresponding ML models generate predictions based on the current activity. These predictions are used as security signals for calculating the overall risk associated with the request.

The RAG-based policy module is implemented separately from the ML models. Security and placement policies are collected and processed into suitable document chunks. These chunks are converted into embeddings and stored in a vector-based knowledge repository. When a user performs an activity that requires policy verification, the system retrieves the most relevant policy content and provides it as context to the language model.

The language model then analyses the retrieved policy together with the user's activity. The resulting response indicates whether the activity is consistent with the defined policy or

represents a potential violation. Since the decision is grounded in retrieved policy information, the system can also provide the relevant policy context supporting the result.

The Explainable AI component is implemented to provide transparency for machine learning predictions. When the security models generate a risk or threat prediction, the XAI mechanism identifies the important input features contributing to that prediction. The explanation is presented along with the prediction so that a security administrator can understand why the system considered the activity normal, suspicious, or high risk.

The Zero Trust decision engine acts as the final control mechanism. It receives the ML predictions, risk assessment, policy-compliance result, user context, and other available security information. Based on these factors, the engine generates an appropriate access decision. A low-risk and policy-compliant request can be allowed, while a suspicious request can require additional verification or be restricted. A high-risk request or confirmed policy violation can result in access being denied.

The placement cell module is implemented as the application layer where the security framework is demonstrated. The module can provide students with access to their authorized placement information and suitable opportunities, while placement coordinators and recruiters receive access according to their authorized responsibilities. The system prevents users from directly accessing protected information without passing through the Zero Trust decision mechanism.

Continuous monitoring is implemented to ensure that an access decision does not permanently establish trust. Subsequent activities are analysed again whenever new security information becomes available. If abnormal behaviour or policy violations are detected after an initially permitted request, the user's risk can be reassessed and the access decision can be changed accordingly.

5.2 MACHINE LEARNING IMPLEMENTATION

The machine learning component consists of three primary models that address different security requirements. The use of multiple models allows the framework to analyse network activity, cloud behaviour, and contextual risk independently.

The Random Forest model is trained using network security data containing features associated with network connections and traffic behaviour. During prediction, the model analyses the processed network attributes and determines whether the observed activity is associated with normal or suspicious behaviour. Its output is forwarded to the risk assessment component.

The Isolation Forest model is trained for cloud anomaly detection. The model learns the general distribution of cloud-related activities and identifies observations that significantly differ from the expected behavioural pattern. Anomalous activities are treated as security signals and contribute to the subsequent risk assessment.

The Gradient Boosting model is used for Zero Trust risk assessment. It analyses relevant contextual and behavioural attributes and generates a risk-related prediction. This prediction is combined with the outputs of the other security models to determine the current security condition of the request.

When multiple machine learning models generate predictions for an access request, their individual risk contributions are combined using a weighted aggregation mechanism. The aggregated machine-learning risk score is calculated as:

$$R_{ML} = \frac{\sum_{i=1}^n w_i R_i}{\sum_{i=1}^n w_i}$$

where R_i represents the risk score generated by model i , and w_i represents the corresponding model weight. The framework assigns weights according to the contribution of each detection model: $w_{\text{NetworkRF}} = 0.25$, $w_{\text{AWSIF}} = 0.20$, $w_{\text{ZTGB}} = 0.30$, and $w_{\text{ZTIF}} = 0.25$. Only models that successfully generate a prediction for the corresponding event contribute to the aggregated ML score.

The aggregated ML risk score is then combined with the contextual security risk to obtain the final risk score:

$$\mathbf{R_{final} = 0.20R_{ML} + 0.80R_{Context}}$$

where R_{ML} represents the normalized machine-learning risk score and R_{Context} represents the contextual security risk derived from the characteristics of the access event. This enables the framework to consider both machine-learning predictions and the operational context of an access request when determining its overall security risk.

The models are evaluated using appropriate performance measures such as precision, recall, F1-score, confusion matrix, and other relevant metrics. Scenario-based testing is also performed because security performance cannot be determined solely from overall accuracy. Different combinations of normal, anomalous, and suspicious activities are evaluated to understand how the integrated framework responds to changing security conditions.

5.3 RAG IMPLEMENTATION

The RAG component is implemented to provide policy-aware security analysis. Policy documents containing access-control, privacy, security, and placement-related rules are collected and converted into machine-readable text. Large documents are divided into smaller meaningful sections so that relevant information can be retrieved efficiently.

The processed policy sections are converted into vector embeddings and stored in a vector-search repository. When a security-related activity occurs, the system creates a query describing the activity and searches the repository for semantically relevant policy sections.

The retrieved policy information is then passed to the language model as contextual information. The model analyses the activity against the retrieved policy and produces a policy-compliance result. This result can indicate whether the request is compliant or represents a violation and can provide the relevant policy context used for the analysis.

The RAG approach also allows policies to be updated without retraining the machine learning models. When an organizational policy changes, the corresponding document can be updated in the knowledge base so that subsequent requests are evaluated against the latest available policy information.

5.4 EXPLAINABLE AI IMPLEMENTATION

The Explainable AI component is implemented to improve the interpretability of the machine learning security predictions. Security decisions can have significant consequences, particularly when a user's access is restricted or denied. Therefore, providing only a prediction without an explanation may not be sufficient for security administrators.

The XAI mechanism analyses the contribution of individual features to the model's prediction. SHAP or LIME can be used to generate local explanations for individual security events. The

resulting information identifies the features that had a significant influence on the predicted risk or threat status.

For example, instead of displaying only “High Risk”, the system can provide an explanation indicating that abnormal network behaviour, unusual cloud activity, or other significant contextual attributes contributed to the increased risk. This makes the security decision easier to interpret and supports further investigation.

The explanations can also be stored together with security events so that administrators can review the reasoning associated with previous decisions.

5.5 ZERO TRUST DECISION IMPLEMENTATION

The Zero Trust decision mechanism integrates the outputs of all major security components. It receives the user's authenticated identity, contextual information, machine learning predictions, risk assessment, RAG policy result, and relevant security explanations.

The decision engine evaluates these inputs before determining the appropriate action. When the request has low risk and satisfies the applicable policies, access can be granted. When the risk is moderate or the available information is insufficient to establish confidence, additional verification can be requested. Suspicious activities can be restricted, while high-risk or confirmed policy-violating requests can be denied.

This implementation ensures that authentication is treated as only one component of the security decision. The user's current behaviour and policy compliance continue to influence access throughout the interaction.

5.6 PLACEMENT CELL ASSISTANCE IMPLEMENTATION

The placement cell assistance module provides the practical application scenario for the proposed framework. The application maintains relevant placement information and provides users with functions according to their authorized roles.

The student interface can provide access to personal placement information and relevant opportunities. The system can compare student attributes with company requirements to

identify potentially suitable opportunities. This assistance is performed while respecting the access permissions defined for the student.

The placement coordinator interface provides authorized administrative functionality, while recruiter-related functionality can be restricted to the information required for recruitment activities. Sensitive information is protected by the Zero Trust decision engine, preventing unauthorized access even when a user has successfully authenticated.

The RAG component can additionally assist in interpreting placement-related policies, such as eligibility rules, application restrictions, or information-access policies. This allows the application to provide assistance while maintaining compliance with predefined organizational rules.

5.7 INTEGRATED IMPLEMENTATION FLOW

The complete implementation follows a sequential security workflow. A user first submits a request through the placement application. The system authenticates the user and collects the relevant contextual and activity information. The information is then preprocessed and passed to the appropriate machine learning models.

The ML models generate threat, anomaly, and risk predictions. At the same time, the RAG component retrieves relevant policies whenever policy verification is required. The Explainable AI mechanism generates an interpretation of the machine learning prediction. These outputs are combined by the Zero Trust decision engine.

The final decision determines whether the requested operation can proceed. If the request is permitted, the corresponding placement service is provided. If additional verification is required, the user is subjected to the appropriate verification mechanism. Suspicious or policy-violating requests are restricted or denied.

The activity is subsequently monitored so that new behavioural information can be incorporated into future decisions. Therefore, the implementation creates a continuous loop of authentication, monitoring, ML analysis, policy verification, explanation, risk assessment, and dynamic access control.

Yes. Since the placement-cell module should be removed, and the VirtualBox environment is being used to demonstrate the security behaviour, Chapter 6 should focus entirely on the actual

Zero Trust project. Since the implementation is still under progress, we should avoid claiming final performance/results that haven't been obtained yet.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 RESULTS

The proposed ZeroTrustAI system integrates Machine Learning, Retrieval-Augmented Generation (RAG), Explainable Artificial Intelligence (XAI), and continuous security monitoring into a unified Zero Trust security framework. The evaluation is carried out in a controlled virtualized environment using VirtualBox. The virtual environment provides an isolated setup in which different user activities, network behaviours, access requests, and security scenarios can be generated and analysed without affecting the host system.

The preprocessing stage handles the heterogeneous datasets used by the different machine learning models. Since the datasets contain different schemas and feature sets, appropriate preprocessing and schema handling are performed before the data is provided to the respective models. This enables the network, cloud, and Zero Trust models to operate with their required input structures. The Random Forest model is used for network threat detection, while the Isolation Forest model identifies anomalous cloud-related behaviour. The Gradient Boosting model is used for Zero Trust risk assessment. The outputs of these models are integrated into the overall security decision process rather than being treated as independent predictions.

The RAG-based policy module analyses user activities against predefined security policies. Relevant policy documents are maintained within a knowledge base, from which appropriate policy information is retrieved for a given request. The retrieved information is provided as context to the language model to determine whether the activity is compliant with the applicable policy or represents a potential violation. The Explainable AI component provides understandable information about machine learning predictions by identifying the major features contributing to a particular threat or risk prediction. This enables security administrators to understand the reasoning behind automated security decisions.

The integrated risk engine was evaluated using ten representative security scenarios covering normal activities, authentication failures, unusual access patterns, suspicious devices, sensitive administrative operations, suspicious network behaviour, and potential data exfiltration. For each scenario, the system generates an ML Score, a Context Score, and a final unified Risk Score. The final score combines the machine-learning assessment with contextual security

information, allowing the system to distinguish between activities that may have similar ML predictions but have different security contexts.

#	Scenario	ML Score	Context	Final Score	Severity
1	Normal Employee Access	66.53	10.00	21.31	LOW
2	Multiple Failed Logins	64.98	44.50	48.60	HIGH
3	Late Night Financial Access	69.97	36.50	43.19	HIGH
4	Unknown Device Download	71.65	71.50	71.53	HIGH
5	Admin Sensitive Delete	67.50	33.50	40.36	HIGH
6	Highly Suspicious Network	72.75	100.00	94.55	CRITICAL
7	Normal Service Account	64.14	1.00	13.63	LOW
8	Possible Data Exfiltration	71.00	86.00	83.00	CRITICAL
9	Normal Manager Activity	67.98	1.00	14.40	LOW
10	Highly Suspicious Login	66.74	100.00	93.35	CRITICAL

Table 6.1 End to End Risk Test Result

The results demonstrate that contextual security information has a significant influence on the final risk assessment. For example, Normal Employee Access produces an ML score of 66.53 but receives a final score of only 21.31 because its contextual risk is low. Similarly, Normal Service Account and Normal Manager Activity produce final scores of 13.63 and 14.40 respectively, resulting in LOW severity. In contrast, Highly Suspicious Network receives a contextual score of 100.00 and a final risk score of 94.55, while Highly Suspicious Login receives a final score of 93.35. Both are therefore classified as CRITICAL.

The Unknown Device Download scenario produces a final score of 71.53 and is classified as HIGH, demonstrating the impact of an untrusted device context. Possible Data Exfiltration also reaches a high final score of 83.00 because of its elevated contextual risk. Overall, the ten test cases result in 3 LOW, 0 MEDIUM, 4 HIGH, and 3 CRITICAL scenarios. These results demonstrate that ZeroTrustAI does not rely only on individual machine learning predictions;

instead, it combines ML outputs with contextual security information to produce a continuous and interpretable risk assessment.

The severity distribution shows that the system successfully differentiates between normal, suspicious, and highly risky activities. The presence of both LOW and CRITICAL classifications across the test scenarios demonstrates the ability of the risk engine to adapt its assessment according to the security conditions associated with each access request.

6.2 SCENARIO ANALYSIS

Scenario-based analysis is being used to evaluate how the proposed Zero Trust framework responds to different security conditions. The scenarios are executed within the VirtualBox environment so that normal and suspicious activities can be safely generated and observed.

In a normal-access scenario, a legitimate user performs an authorized operation without exhibiting abnormal behaviour. The security models are expected to identify the activity as normal, while the RAG component verifies that the activity complies with the relevant policy. The Zero Trust decision engine can consequently permit the request.

In a suspicious-behaviour scenario, the user generates activity that differs from the expected behavioural pattern. Such activity may result in a network threat prediction, cloud anomaly, or increased risk score. The system then reassesses the request and can require additional verification or restrict access according to the resulting security condition.

In a policy-violation scenario, an activity may not necessarily appear as a network attack but may still violate an organizational security rule. The RAG component retrieves the relevant policy and analyses the activity against it. If a violation is identified, the result is forwarded to the Zero Trust decision engine for an appropriate response.

A combined-risk scenario is also considered in which multiple security indicators occur simultaneously. For example, abnormal network behaviour may occur together with unusual cloud activity and an attempt to access a restricted resource. The combined security signals can increase the overall risk and lead to a stronger Zero Trust response.

6.3 EXPLAINABLE SECURITY DECISIONS

Explainable Artificial Intelligence is incorporated into the proposed framework to improve the transparency of machine learning-based security decisions. A conventional machine learning model may indicate that an activity is suspicious without providing sufficient information about the factors responsible for the prediction. The proposed XAI layer is intended to address this limitation by identifying the features that contribute significantly to the prediction.

For example, when an activity receives a high-risk classification, the system can provide information indicating that abnormal network behaviour, unusual cloud activity, or other relevant attributes contributed to the increased risk. This information can assist administrators in investigating the event and understanding the reason for the security decision.

The proposed framework also combines XAI with RAG-based policy analysis. XAI explains why the ML model produced a particular prediction, while the RAG component identifies which policy is relevant to the activity and whether the activity complies with that policy. This combination is expected to provide a more transparent security decision.

6.4 VIRTUALIZED SECURITY ENVIRONMENT

The proposed system is being evaluated within a controlled virtualized environment created using VirtualBox. Virtualization provides an isolated environment for executing the application and generating security-related activities without directly affecting the host operating system. This environment is useful for demonstrating the behaviour of the Zero Trust framework under different security scenarios.

The virtual environment can contain the application and required security components while simulated users and activities are used to generate requests. Network communication, authentication events, resource-access attempts, and other relevant activities can be observed within the environment. These activities can subsequently be supplied to the security analysis pipeline.

The use of VirtualBox also allows different security scenarios to be reproduced consistently. Normal activities can first be established as a baseline, after which abnormal or suspicious activities can be introduced to observe how the machine learning models and Zero Trust decision mechanism respond.

The virtualized environment therefore acts as a controlled test platform for validating the interaction between the application, security monitoring components, machine learning models, RAG policy analysis, Explainable AI, and the Zero Trust decision engine.

Since the project is currently under development, the VirtualBox-based experimental environment is being progressively integrated with the remaining security components. Final observations and quantitative evaluation results will be included after completion of the implementation.

6.5 DISCUSSION

The proposed framework is designed to combine multiple security mechanisms rather than relying on a single detection technique. Machine learning provides the ability to identify threats, anomalies, and behavioural risks, while Zero Trust provides the architecture through which these security signals can influence access decisions.

The RAG component introduces policy awareness into the framework. This is important because an activity does not necessarily have to be a technical attack to represent a security concern. An otherwise normal activity can still violate an organizational policy, and the RAG mechanism provides a method for identifying such violations using the relevant policy documents.

Explainable AI complements the ML models by making their predictions easier to interpret. This is particularly important when the system produces a high-risk prediction or restricts a request. The administrator can use the explanation to understand the major factors contributing to the prediction.

The VirtualBox environment provides a controlled platform for testing these interactions. Different security conditions can be reproduced and the resulting system behaviour can be observed without requiring a production environment.

As the implementation progresses, the final evaluation will focus on the accuracy and reliability of the individual ML models as well as the effectiveness of the complete Zero Trust decision process. The analysis will consider whether the framework can correctly identify suspicious behaviour, detect policy violations, generate meaningful explanations, and produce appropriate access decisions.

6.6 PERFORMANCE EVALUATION

The machine learning models will be evaluated using appropriate performance metrics such as precision, recall, F1-score, confusion matrix, and accuracy where applicable. For the anomaly-detection component, the evaluation will focus on the model's ability to identify activities that differ from expected behaviour.

However, model accuracy alone will not be considered sufficient for evaluating the proposed security framework. The system will also be evaluated using different security scenarios within the VirtualBox environment. These scenarios will examine how the framework responds to normal requests, anomalous behaviour, suspicious activities, and policy violations.

The final evaluation will therefore consider both individual model performance and overall Zero Trust decision behaviour. The objective is to determine whether the integrated system can effectively combine ML predictions, policy information, XAI explanations, and contextual security information to generate appropriate access decisions.

Since implementation is currently under progress, the final numerical results will be added after completion of the remaining experiments.

CHAPTER 7

CONCLUSION AND FUTURE WORKS

7.1 CONCLUSION

The proposed ZeroTrustAI framework integrates Zero Trust Architecture, Machine Learning, RAG, XAI, and continuous security monitoring to provide adaptive and context-aware cybersecurity risk assessment. Random Forest, Isolation Forest, and Gradient Boosting support threat classification, anomaly detection, and risk assessment, while RAG enables policy-aware analysis and XAI improves the transparency of security decisions. The framework brings these components together to provide a unified approach for identifying security risks and supporting informed access-control decisions. The Zero Trust Decision Engine combines these security insights to continuously evaluate access requests and classify them as ALLOW, VERIFY, RESTRICT, or DENY. This approach reduces reliance on static authentication and supports continuous verification based on user behaviour, security context, detected anomalies, and policy compliance. The proposed architecture provides a foundation for improving organizational security by adapting access decisions according to changing risk conditions and security requirements.

7.2 FUTURE WORKS

Future work will focus on developing ZeroTrustAI into a low-code Zero Trust security platform that enables organizations to configure security policies, connect data sources, and create security workflows with minimal programming. Visual components can be provided for threat detection, anomaly detection, risk assessment, RAG-based policy verification, XAI, and access decisions. This will make the framework more accessible to organizations that require security automation without extensive development effort. The platform can further be enhanced with real-time monitoring, adaptive machine learning, threat-intelligence integration, automated response, enterprise-scale deployment, and advanced policy management. Larger datasets and diverse security scenarios will also be used to evaluate detection performance, risk assessment, scalability, and overall system effectiveness. Future versions can additionally support flexible integration with different organizational environments and security infrastructures, improving the practical applicability of the proposed system.

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ZeroTrustAI: A Machine Learning-Driven Framework for Continuous Cybersecurity Risk Assessment

Mahesmeena K.

*Department of Computer Science
and Engineering*

Rajalakshmi Engineering College
Chennai, India

mahesmeena.k@rajalakshmi.edu.in

Monish D.Y.

*Department of Computer Science
and Engineering*

Rajalakshmi Engineering College
Chennai, India

230701195@rajalakshmi.edu.in

Manicka Meenakshi S.

*Department of Computer Science
and Engineering*

Rajalakshmi Engineering College
Chennai, India

230701173@rajalakshmi.edu.in

Lavanya R.

*Department of Computer Science
and Engineering*

Rajalakshmi Engineering College
Chennai, India

230701162@rajalakshmi.edu.in

Mohanapriya R.

*Department of Computer Science
and Engineering*

Rajalakshmi Engineering College
Chennai, India

230701191@rajalakshmi.edu.in

Abstract—The rapid adoption of cloud technologies has led to a series of cybersecurity problems and exposed weaknesses in traditional security models. Traditional access-control methods rely on initial authentication and fixed trust, so they are less effective against behavioral changes, unexpected activity, and rule violations. This paper offers a machine learning-based Zero Trust architecture that combines machine learning, Retrieval-Augmented Generation, Explainable Artificial Intelligence, and continuous security monitoring. The architecture keeps on checking permissions using behavioral and contextual elements instead of granting constant trust to the previously verified user. A Random Forest algorithm is used for classification of attacks, and Isolation Forest for the detection of anomalies within the cloud activities. Gradient Boosting is used for assessing risks. Meanwhile, there is a policy analysis mechanism based on Retrieval-Augmented Generation technology for searching for needed policies from outside sources, while Explainable Artificial Intelligence enables to give understandable reasoning regarding the security decisions made with the help of Machine Learning technology. The Zero Trust Decision Engine separates incoming requests into ALLOW, VERIFY, RESTRICT, or DENY categories based on processed risk.

Index Terms—Zero Trust Architecture, Cybersecurity, Machine Learning, Anomaly Detection, Retrieval-Augmented Generation, Explainable Artificial Intelligence

I. INTRODUCTION

With the rapid development of cloud computing, web apps, remote access services, and distributed digital solutions, the cybersecurity environment today is completely different compared to previous years. Nowadays, organizations use various devices, networks, cloud systems, and other remote locations to access their resources, which increases the types of threats, such as compromised credentials, malicious use, insider attacks, illegal access, and violations of rules. Consequently,

conventional perimeter-based security models can no longer provide sufficient protection to digital resources in the modern world.

Zero Trust Architecture represents an alternative to current approaches by adopting continuous verification practices. Unlike current approaches that treat validated user/device as trusted, Zero Trust approaches every request according to its identity, device status, context of the request, sensitivity of the resource, and other security signals.

Zero Trust security can be enhanced by using machine learning to identify undesirable and unusual events using behavioral as well as security information. This framework implements the Random Forest technique for classifying risks of network attacks, the Isolation Forest technique for detecting anomalies occurring in the cloud, and the Gradient Boosting technique for assessing risks related to Zero Trust. They generate additional security signals that help evaluate various types of requests for access.

Another notable challenge faced is compliance with the security policy applied by the organization, which is normally introduced as a collection of documents defining the rules related to authentication, authorization, data access, privacy, and usage of the resources. Manual verification of the policy is ineffective; that is why the unique solution provided here is Retrieval-Augmented Generation (RAG) that is responsible for retrieving policies from external sources to offer them as evidence when analyzing violations within the policies.

Moreover, it is not easy for decision-makers to understand artificial intelligence. Explainable AI is applied to determine the most significant elements impacting the process of predicting threats and risks and to increase the level of transparency

of decisions made in the area of security. The suggested framework provides dominant means of distinguishing threats, assessing risk, verifying the policy, applying Explainable AI, and continuously monitoring the situation based on the Zero Trust principle.

The effectiveness of the framework is assessed in the controlled VirtualBox environment using normal and abnormal, suspicious, and policy-violating scenarios. The study aims to establish an adaptive, explainable, policy-aware, and context-sensitive access control system that responds to changes in the security environment.

II. LITERATURE REVIEW AND RESEARCH GAP

A. Literature Review

With the emergence of cyber threats, Zero Trust Architecture (ZTA) has become a valuable security solution for distributed, hybrid, and cloud scenarios. According to NIST, Zero Trust assumes that access to users and devices does not depend on their position in the network. In this regard, a user's identity, device, and context are analyzed to ensure that access is granted continuously [1], [2].

Studies have examined the synergistic use of Machine Learning (ML) technologies in Zero Trust-based security systems. Goshikonda and Pulipati developed a framework that combines ML algorithms, behavioral analytics, dynamic access control, and adaptive risk scoring [3]. Another example is SmartTrust presented by Lilhore et al. [4], in which CNN, LSTM, reinforcement learning, and transformers were implemented to detect threats in cloud services based on Zero Trust. Context-aware behavioral analysis was developed by Zhang and Wu [5], while Folino et al. created models for anomaly detection based on an ensemble approach [6]. Additionally, the solutions proposed by Manoharan et al. and Sarhan and Altwaijry prove the efficiency of supervised ML technology in the insider threat detection process [7], [8]. However, all the mentioned studies deal with anomaly detection rather than linking behavioral intelligence with continuous authorization in practice.

Dynamic access control has been examined as a method for adjusting permissions based on changing trust and risk conditions. Huang et al. introduced a trust-driven access control methodology for cloud-based environments [9]. On the other hand, Wang et al. developed a Zero Trust-based technique for managing access to cloud-based infrastructures [10]. Nevertheless, integration with respect to machine learning (ML)-based risk assessment is still not widely done.

Explainable AI (XAI) technology can help improve ML transparency in cybersecurity solutions. For example, Gaspar et al. and Sharma et al. used LIME and SHAP for making predictions regarding cyber intrusion detection [11], [12]. However, these works mostly focus on explanations for singular detection models and do not consider Zero Trust-based decisions.

Retrieval-Augmented Generation (RAG) enables security systems to achieve dynamic retrieval of recent organizational policies and security intelligence. For example, Lewis et

al. discussed RAG architecture [13]. Adediran et al. then applied RAG in conjunction with AWS CloudTrail and MITRE ATT&CK for cloud-based threat detection [14]. MITRE ATT&CK also provides valuable threat data that can be used for identifying effective techniques deployed by malicious users [15]–[17].

Commercial platforms currently offer a range of capabilities. Microsoft Defender for Identity can detect identity threats and analyze behavior. At the same time, Splunk Enterprise Security has risk-based security analytics, and CrowdStrike Falcon Identity Protection has behavior baselining and risk-based identity protection. The ZeroTrustAI framework does not intend to present the absence of ML, behavior analytics, or Zero Trust from these existing platforms; rather, its distinctiveness is to integrate these features among themselves within a common decision-making process.

B. Research Gap

Research has shown that zero-trust architecture, machine learning-based threat detection, combined access control, explainable AI, and retrieval-augmented generation work effectively as separate security mechanisms, but their potential is not exploited through their possible integration into a single, continuously working access-control framework. In particular, existing zero trust and machine learning methods mainly deal with threat detection and risk assessment tasks, while studies applying explainable AI focus on explanation of the results obtained by certain models. Similarly, retrieval-augmented generation technologies are primarily used for threat intelligence and knowledge retrieval tasks, instead of actual policy verification in the process of access decisions.

As a result, there is still a lack of a holistic Zero Trust framework for incorporating different types of ML security signals in the process of continuous risk assessment, the use of RAG for policy verification, as well as practical application of XAI for security reasoning. The ZeroTrustAI framework aims to fill this gap by bringing all of the aforementioned elements together. The unique feature of the ZeroTrustAI framework is that rather than relying on a single ML model for making access decisions, it relies on attack classification, anomaly detection, risk assessment, policy verification, and explainability for determining whether to ALLOW, VERIFY, RESTRICT, or DENY access.

III. PROPOSED FRAMEWORK

The suggested structure provides adaptive and contextual access control by employing Zero Trust Architecture, Machine Learning, Retrieval-Augmented Generation (RAG), Explainable Artificial Intelligence (XAI), and continuous security monitoring. Every request for access is evaluated using identity, device, network, cloud, behavioural, contextual, and legal information, instead of the limited criteria of authentication or location in the network.

A. Framework Architecture

The framework described includes gathering context, pre-processing data, analyzing security using machine learning,

carrying out risk evaluation, confirming policies in accordance with the RAG model, applying Explainable Artificial Intelligence (XAI), making the judgment, and ongoing monitoring of activities. The moment a person or device applies for access to a system resource, authentication and contextual information, including the status of the device, details of the applicant's request, network behavior, and cloud operations, are obtained. Authentication alone is not enough to confirm trust.

Security information received from different sources undergoes the stage of pre-processing and schema management, including cleaning of data, dealing with losses, transforming important data, converting numerical information, normalizing it where it is needed, and aligning schema. The information being processed is then received by the appropriate ML model.

B. ML-Based Security Analysis

Various ML models are applied for various security needs. Random Forest takes care of network threat classification, while Isolation Forest checks for abnormal activities in the cloud environment. Gradient Boosting ensures Zero Trust risk evaluation based on any relevant security and situational signals it can find. The results coming from the mentioned models are put together as complementary indicators, rather than being viewed as separate access decisions.

C. Risk-Based Zero Trust Assessment

The system constantly evaluates risks linked to an access request using further prediction technologies, some behavioral data, contextual characteristics, and policy compliance. Considering the overall security situation, low-risk and compliant requests obtain ALLOW, moderate or unsure ones are given VERIFY, suspicious acts receive RESTRICT, and high-risk requests or definite policy violations are assigned DENY. Thus, access decisions can be made according to the changing security situation.

D. RAG-Based Policy Verification

Retrieval-Augmented Generation (RAG) is employed to validate the compliance of events with the security rules of an organization. Security policy documentation that comprises provisions regarding authentication, authorization, security of information, privacy, as well as usage of resources is stored outside the organization. When an access request is received, relevant data is retrieved and input into the language model. After that, the request is measured against the retrieved information for compliance. Since the policy documents are maintained separately from the machine learning models, any change in the policies can be applied without updating the threat detection algorithms.

E. Explainable AI

The concept of Explainable Artificial Intelligence (XAI) makes it simpler to understand the comprehensive nature of decisions made with the help of ML in the field of security. Methods like SHAP and LIME effectively present the key parameters leading to the classification of a threat or prediction

of the risk. The conclusion received from the above-mentioned measures, together with information regarding relevant policies taken by RAG, will give the administrator the possibility to understand the reasons as well as the policies affecting the decision that has been made.

F. Unified Decision Engine and Continuous Monitoring

Zero Trust decision engine brings together the parts of authentication, contextual information, ML outputs, risk assessment, RAG policy outcomes, and XAI information to make the final access decision. No one ML model solely determines access. After access is granted, continuous monitoring of access continuously captures actions of users, devices, networks, and resources in the system. Any case of suspicious actions, anomalous cloud activities, or policy violations may lead to the reassessment of the request, and the state can be changed from ALLOW to VERIFY, RESTRICT, or DENY in the future.

The framework is analysed in a controlled environment employing VirtualBox in such a manner as to create typical, atypical, suspicious, and non-compliant conditions for simulating the comprehensive operation of the framework and demonstrating its capability to integrate detection of threats, risk assessment, verification of compliance with policies, explanation of the results, and implementation of constant monitoring into adaptive Zero Trust access control.

IV. SYSTEM IMPLEMENTATION AND EXPERIMENTAL SETUP

A. Data Collection and Preprocessing

The presented methodology integrates security data obtained from various sources in order to signify various components safeguarding an access request. For the industry tests, the CICIDS network intrusion dataset is used along with the application logs, authentication logs, SQLite database activity, fabricated cloud security events, as well as policies of the business organization. All of the provided sources represent information regarding the network, user identity, applications, cloud storage, as well as organizational policies needed for ZeroTrustAI-based decision-making.

As the investigated data is made up of different formats and feature representations, the preprocessing is performed prior to making a machine learning analysis. Thus, duplicates and wrongly obtained records are eliminated, whereas missing data is treated in accordance with the specifications of the concrete dataset.

Categorical variables have been changed into numerical values that can be used efficiently in machine learning models while numeric attributes are standardized when needed. Once data is processed, it gets structured according to the kind of security task solved by a particular model. Features associated with network problems are used in the Random Forest model, cloud and activity events are processed in Isolation Forest model, and variable features are applied to the Gradient Boosting risk model. The generated outputs are then sent to the dynamic risk engine for comprehensive risk assessment.

ZeroTrust AI – Zero Trust Access Monitoring & Analytics System

AI/ML Powered Risk Assessment, RAG Context & Zero Trust Decision

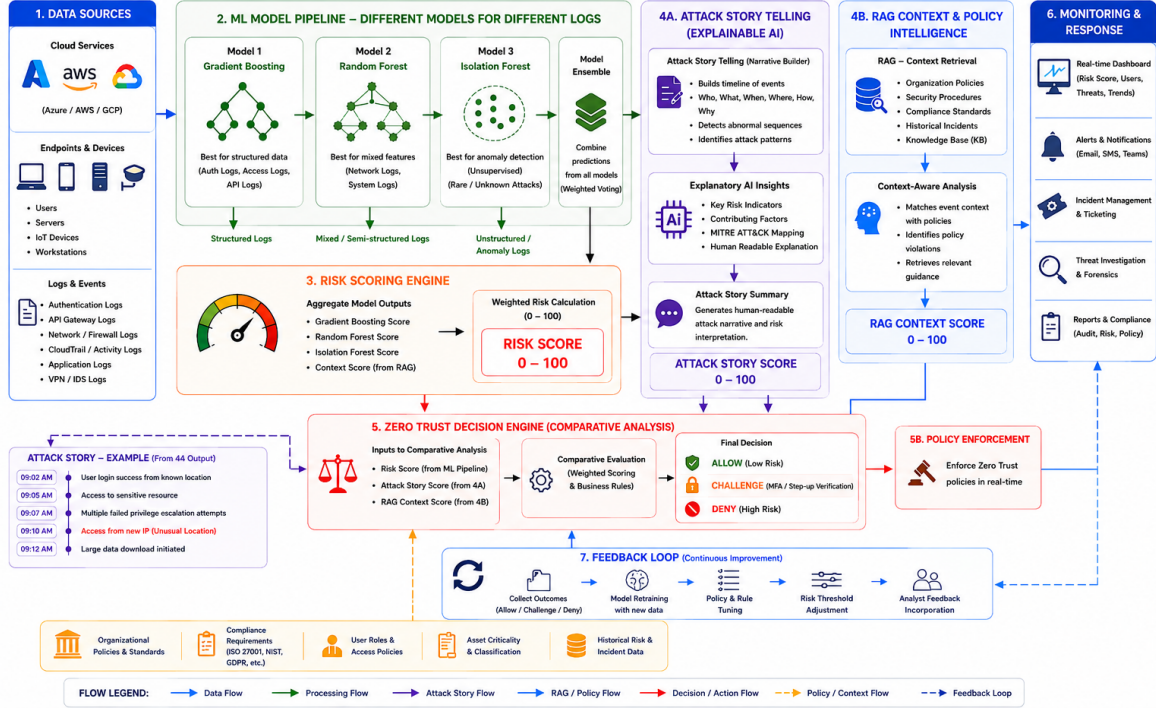


Fig. 1. System architecture of the proposed framework

B. Machine Learning Models and Pipeline

This system uses many ML models, each performing a specific function of security analysis. Random Forest works for classifying known attack types in a supervised manner, and Isolation Forest serves the purpose of anomaly detection for logs and events from the cloud and node. Specific combinations of Gradient Boosting decide how much security risk is present from behavioral characteristics and features. Results from the system's models are treated as additional means of security signal processing and are sent to the risk processing unit instead of being used separately to grant access.

C. Dynamic Risk Scoring and Severity Classification

To operationalize continuous risk assessment within the proposed Zero Trust AI framework, a dynamic risk-scoring mechanism is employed that integrates machine-learning-based threat inference with contextual security intelligence. Rather than relying solely on the output of an individual detection model, the proposed approach synthesizes heterogeneous behavioral and contextual signals into a unified 0–100 risk score.

The final risk score is formulated as:

$$R_{\text{final}} = 0.20R_{\text{ML}} + 0.80R_{\text{Context}}$$

where R_{ML} represents the normalized machine-learning risk score and R_{Context} represents the contextual security risk derived from the characteristics of the access event.

1) Machine-Learning Risk Aggregation: When multiple trained detection models are applicable to an event, their individual risk contributions are aggregated through a weighted normalization mechanism:

$$R_{\text{ML}} = \frac{\sum_{i=1}^n w_i R_i}{\sum_{i=1}^n w_i}$$

where R_i denotes the risk score generated by model i , and w_i denotes its corresponding model weight. The framework assigns weights according to the configured contribution of each detection model:

$$\begin{aligned} w_{\text{NetworkRF}} &= 0.25 \\ w_{\text{AWSIF}} &= 0.20 \\ w_{\text{ZTGB}} &= 0.30 \\ w_{\text{ZTIF}} &= 0.25 \end{aligned}$$

Only models that successfully generate a prediction for the corresponding event contribute to the aggregated ML score, thereby preventing unavailable models from artificially reducing the resulting risk.

2) Context-Aware Risk Computation: The contextual component captures security-relevant characteristics that may not be completely represented by the ML prediction. These include

authentication failures, repeated login attempts, IP reputation, unusual access time, device trustworthiness, sensitive actions and resources, privileged roles, network volume, encryption status, and session duration.

The contextual risk is bounded to the operational range:

$$R_{\text{Context}} = \min \left(100, \sum_{j=1}^m C_j \right)$$

where C_j represents the contribution of an individual contextual security indicator.

This design enables the system to distinguish between events that may receive similar ML predictions but exhibit substantially different operational security contexts.

3) *Risk Severity Classification*: The resulting unified risk score is mapped into four operational severity levels:

$$\text{Severity}(R) = \begin{cases} \text{LOW}, & 0 \leq R < 25 \\ \text{MEDIUM}, & 25 \leq R < 40 \\ \text{HIGH}, & 40 \leq R < 75 \\ \text{CRITICAL}, & 75 \leq R \leq 100 \end{cases}$$

Consequently, the framework transforms heterogeneous access telemetry and ML predictions into an interpretable security decision that can be consumed by monitoring dashboards, alerting mechanisms, and subsequent access-control workflows.

The proposed risk engine therefore establishes a context-aware, model-agnostic risk quantification layer that bridges probabilistic machine-learning inference and operational Zero Trust decision-making, enabling continuous and interpretable assessment of access risk rather than relying on isolated binary anomaly predictions.

D. RAG-Based Policy Analysis

Through the use of the RAG component, the platform has the possibility of reviewing security incidents by utilizing internal policies and regulations. To achieve this objective, the policies are converted into a machine-readable format, segmented into different fragments for easier storage, and included in a knowledge base searchable by using vector information. When a security incident occurs and requires verification of policy compliance, the platform is able to search for relevant information and use it as the background for its examination. Among the advantages of the presented approach, one of the most significant is that there is no need for retraining of models in case of policy revision.

E. Attack Story Reconstruction and Explainable AI

The purpose of an attack story is to link similar incidents together and create a coherent sequence of events. Through time of occurrence, username, session, and sequence information, administrators can track the development of any

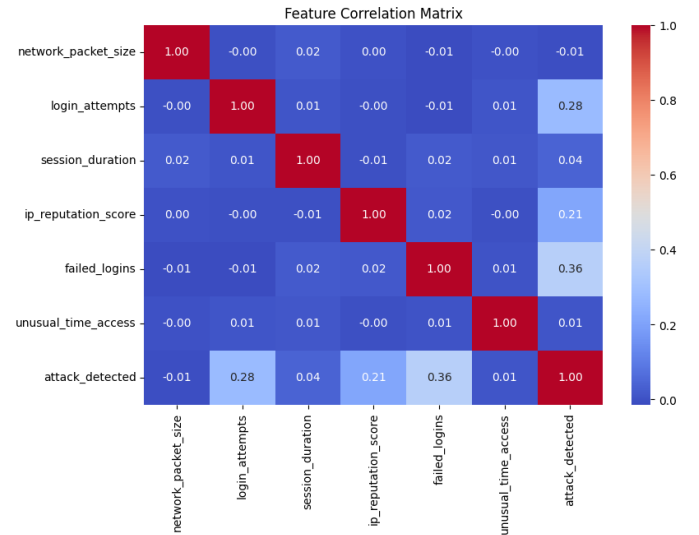


Fig. 2. Feature correlation matrix of security attributes used for attack detection

attacks. Explainable AI offers another level of transparency by identifying the various components that make up a machine learning prediction. Rather than providing just a risk level, the system will also indicate if it was caused by abnormal activity on the network, unusual cloud behaviour, etc.

V. RESULTS AND DISCUSSIONS

A. Experimental Results

Experiments were conducted to evaluate the ability of the proposed ZeroTrustAI framework to detect anomalous activities and quantify security risk under different access conditions. The evaluation considers multiple security attributes, including packet size, login frequency, session duration, IP reputation score, failed login attempts, and access time. The correlation analysis shown in Fig. 2 indicates that attack-related activity is influenced by multiple security attributes rather than a single feature. Among the analyzed attributes, failed login attempts exhibit the highest correlation with attack detection (0.36), followed by login frequency (0.284) and IP reputation score (0.218). The relatively low individual correlation values indicate that multiple security signals should be considered jointly for comprehensive risk assessment.

End-to-end tests were subsequently conducted using representative access scenarios to evaluate the behavior of the proposed risk engine under different security conditions. The results are presented in Table I. The ML-based risk scores remain within a relatively narrow range, while contextual security conditions substantially influence the final risk score. This demonstrates the role of contextual information in differentiating access requests that may have similar machine-learning risk estimates.

The results demonstrate that the centralized risk engine combines security signals from multiple analytical components, including network-based predictions, cloud security analysis, Gradient Boosting risk estimation, and Isolation

TABLE I
ZEROTrUSTAI END-TO-END RISK TEST RESULTS

#	Scenario	ML Score	Context	Final Score	Severity
1	Normal Employee Access	66.53	10.00	21.31	Low
2	Multiple Failed Logins	64.98	44.50	48.60	High
3	Late Night Financial Access	69.97	36.50	43.19	High
4	Unknown Device Download	71.65	71.50	71.53	High
5	Admin Sensitive Delete	67.50	33.50	40.36	High
6	Highly Suspicious Network	72.75	100.00	94.55	Critical
7	Normal Service Account	64.14	1.00	13.63	Low
8	Possible Data Exfiltration	71.00	86.00	83.00	Critical
9	Normal Manager Activity	67.98	1.00	14.40	Low
10	Highly Suspicious Login	66.74	100.00	93.35	Critical

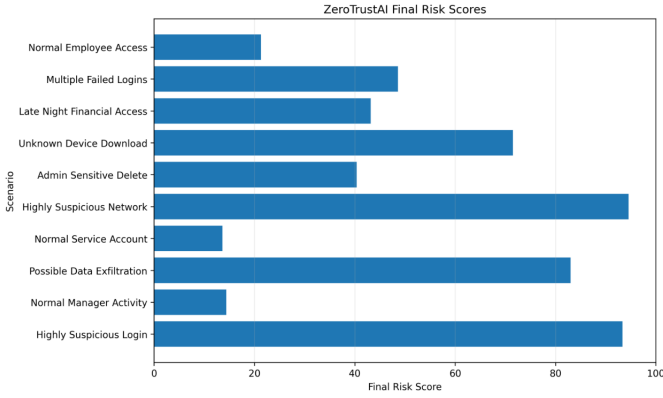


Fig. 3. Risk score variation across different security scenarios

Forest anomaly detection. These outputs are integrated with contextual security information to generate a unified risk score. The contextual component therefore has a significant influence on the final risk assessment, allowing the framework to distinguish between normal activities and scenarios involving suspicious devices, abnormal access patterns, sensitive operations, or potentially malicious network behavior.

As illustrated in Fig. 3, normal activities such as normal employee access, normal service-account activity, and normal manager activity produce relatively low final risk scores of 21.31, 13.63, and 14.40, respectively. In contrast, scenarios involving possible data exfiltration, highly suspicious network activity, and highly suspicious login behavior produce substantially higher scores of 83.00, 94.55, and 93.35, respectively, and are classified as Critical. These results demonstrate that the proposed framework represents security posture as a continuously varying risk state rather than as a simple binary attack or non-attack outcome. Such continuous risk assessment supports Zero Trust access control by allowing security decisions to adapt according to changes in user behavior, device conditions, contextual attributes, and other security signals.

VI. LIMITATIONS AND FUTURE SCOPE

The limitations of the current project arise from its dependence on existing datasets, application logs, and organizationally controlled safety scenarios, which cannot entirely exhibit the degree of complexity and size of big enterprises. The success of machine learning solutions depends on the accuracy and amount of security data accessible; working with different types of data sources as well as processing a large number of events is difficult. Future studies will focus on converting the suggested system into a Low-Code Zero Trust Security Platform, which will allow users to connect various sources of security data, cloud services, applications, and databases with little programming drag. The developed platform gives visual workflows for setting up ML models, security policies, risk-scoring processes, and automated Zero Trust actions, making the system more flexible, scalable, and easier to implement in real organizations.

VII. CONCLUSION

The proposed study shows a Machine Learning-based Zero Trust security framework that integrates Zero Trust Architecture with Machine Learning, RAG, Explainable AI, and continuous security monitoring to resolve the issues found in conventional perimeter-based security. Random Forest, Isolation Forest, and Gradient Boosting perform network threat classification, cloud anomaly detection, and risk evaluation, respectively. With RAG, access activities can be compared with applicable and updatable security policies, while XAI enhances transparency of the decisions made by the ML-based security framework.

The framework integrates the security signals via a single Zero Trust decision mechanism that makes a real-time classification of requests as ALLOW, VERIFY, RESTRICT, or DENY. Continuous monitoring makes it possible for the system to review access once the conditions of the user, device, or activity change. A controlled VirtualBox environment is a safe environment in which normal, anomalous, suspicious,

and policy-violating situations can be tested. In a nutshell, the proposed framework should be a solution that would allow providing adaptive, policy-aware, explainable, and context-sensitive access control, thus boosting the security of modern cloud and distributed environments.

STATEMENTS AND DECLARATIONS

Funding

No funding was received for conducting this study.

Data Availability

This research makes use of the publicly available datasets on cybersecurity and simulated logs of cloud activities. The data processing and analysis techniques described in this paper are reproducible by anyone with access to appropriate publicly available datasets and the implementation information described in the paper. No personally identifiable or confidential organizational data were used in this study.

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Submission Summary

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International Conference on computing,communication and wireless Networks

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Paper Title

ZeroTrustAI: A Machine Learning-Driven Framework for Continuous Cybersecurity Risk Assessment

Abstract

The rapid adoption of cloud technologies has led to a series of cybersecurity problems and exposed weaknesses in traditional security models. Traditional access-control methods rely on initial authentication and fixed trust, so they are less effective against behavioral changes, unexpected activity, and rule violations.

This paper offers a machine learning-based Zero Trust architecture that combines machine learning, Retrieval-Augmented Generation, Explainable Artificial Intelligence, and continuous security monitoring. The architecture keeps on checking permissions using behavioral and contextual elements instead of granting constant trust to the previously verified user.

A Random Forest algorithm is used for classification of attacks, and Isolation Forest for the detection of anomalies within the cloud activities. Gradient Boosting is used for assessing risks. Meanwhile, there is a policy analysis mechanism based on Retrieval-Augmented Generation technology for searching for needed policies from outside sources, while Explainable Artificial Intelligence enables to give understandable reasoning regarding the security decisions made with the help of Machine Learning technology.

The Zero Trust Decision Engine separates incoming requests into ALLOW, VERIFY, RESTRICT, or DENY categories based on processed risk.

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Authors

Monish D Y (Rajalakshmi Engineering College) <230701195@rajalakshmi.edu.in>

Manicka Meenakshi S (Rajalakshmi Engineering College) <230701173@rajalakshmi.edu.in>

Lavanya R (Rajalakshmi Engineering College) <230701162@rajalakshmi.edu.in>

Mohanapriya R (Rajalakshmi Engineering College) <230701191@rajalakshmi.edu.in>

Mahesmeena K (Rajalakshmi Engineering College) <mahesmeena.k@rajalakshmi.edu.in>

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